Part Note Codes, Part Traceability, and Product-Unit Records

Which items must be serialized or lot/batch recorded, why, how far the records cascade, and which part note codes put those requirements on a parts list.

TRAINEE HANDOUT

REVISION	C
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CONTENTS	8 modules · 51 sections
NAME	
DATE	

Work through the sections in order. Each ends with a question that checks the point it made. Each module then has a parts-list drill: read a line, pick the codes, get instant feedback. The written exercises come last. They are what your trainer will actually read.

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Glossary

Who it is for. Engineers who apply part note codes, and everyone who acts on them: planners, buyers, inspectors, build technicians, and configuration management staff. No drawing experience is assumed.

Prerequisites. None. Work the modules in order. Each module builds on the one before it.

Time. Allow 30 to 45 minutes for each module, plus the drill and the exercises. Few people finish in one sitting, and the course is not designed for that.

MODULE 1

Why traceability, and the eight codes

Why traceability matters, how to read a parts list line, the eight codes, and who owns each requirement source

M1-01 Why part traceability exists

WHAT YOU WILL BE ABLE TO DO

Explain what a serial number and a lot/batch number each tell you.

Sort the eight part note codes into identity, link, and condition codes.

Name who imposes the codes, who captures the records, who reconciles them, and who owns each requirement source.

Use the four questions on one parts list line.

A turbine disk cracks in service. The investigation traces the crack to an inclusion in the forging. The next question is simple. Which other disks came from that heat of material, and which aircraft are they on?

Part traceability records answer that question. Without them, the answer is every disk in the fleet.

Two identifiers do the work. A **serial number** says which unit. A **lot/batch number** says which group of units shared a material, a process run, or a test. A traceability record then ties each unit to the assembly it went into.

Traceability turns a fleet-wide problem into a list of tail numbers.

The list is only as good as the records behind it. One missing link breaks the chain of records at that point, and everything above the break is unknown.

These are **product-unit records**. They describe individual units and lots as built. The design says what each item must be. The product-unit records say what each delivered unit actually contains.

THE SAME RECORDS SERVE MORE THAN RECALLS

Each unit counts its own remaining life. A life-limited part without a serial number cannot be retired on time.

A test result that qualifies one part must stay with that part. When a supplier reports a suspect lot, the program must find that lot wherever it went.

Each of these needs the same thing: an identifier on the item, and a record of where the item went.

CHECK YOURSELF · M1-01

A forging supplier reports that one heat of titanium had a chemistry nonconformance. What do part traceability records let the program do?

- [ ] **A** Replace the material design values for that alloy across all programs
- [ ] **B** Nothing until the drawing is revised to call out a different heat
- [ ] **C** Identify the units made from that heat and the assemblies they went into
- [ ] **D** Show that the heat met specification, since it passed receiving inspection

M1-02 Reading one parts list line

A parts list names every item in an assembly. Each line is one item. Here is one line from a main landing gear parts list.

Find no.	Part number	Nomenclature	Qty	Notes
4	EX-5512-300	STRUT, SHOCK, MAIN LANDING GEAR	1	SER, LL, TR

Read the line from left to right.

- The **find number** (Find no.) ties the line to the same item on the drawing picture.
- The **part number** identifies the design.
- The **nomenclature** names the item.
- The **quantity** (Qty) says how many go into one assembly.
- The **notes** carry the part note codes. Each code points to a note text with the full instruction.

This strut carries three codes. **SER** says each strut gets its own serial number. **LL** says the strut has a life limit. **TR** says the strut has children that carry their own serial or lot/batch numbers. The records must link those children to this strut.

Each code on the line has a reason. The strut wears out by landing cycles, so its life counts per unit. Its seals carry lot/batch numbers, so the records must show which seals went into which strut. The rest of this course teaches those reasons.

CHECK YOURSELF · M1-02

On the strut line, which code says the records must show which seals went into this strut?

- [ ] **A** TR
- [ ] **B** SER
- [ ] **C** LL
- [ ] **D** The part number

M1-03 The eight part note codes

The strut line used three codes. There are eight in all. Learn each one by what it drives.

Code	Name	What the code drives
SER	Serialization Recording Required	Record serial numbers, or apply and record serial numbers
LOT	Lot/Batch Recording Required	Record lot/batch numbers, or apply and record lot/batch numbers
TR	Traceability Records Required	Generate the parent-child traceability records of items in the product tree
IUID	Item Unique Identification Required	Mark the item with a unique item identifier and report it to the DoD IUID Registry
LL	Life Limited	Identify the item as life-limited and state its limit
SH	Special Handling Required	State the handling, packaging, or environmental control required
CSI	Critical Safety Item	Identify the item as a Critical Safety Item
FC	Fracture-Critical Item	Identify the item as a Fracture-Critical Traceable Part or a Fracture-Critical Part

A part note code never travels alone. It goes on the parts list with its **note text**. The code is the short handle. The note text is the instruction the build environment reads.

Here is one, as it appears on a parts list:

SER SERIALIZATION RECORDING REQUIRED. THIS ITEM IS SERIALIZED AND TRACEABILITY MUST BE MAINTAINED BETWEEN THIS ITEM AND ALL HIGHER-LEVEL PHYSICAL ITEMS UP TO THE END ITEM.

The rest of this course teaches which items take which codes, and why.

CHECK YOURSELF · M1-03

Which part note code directs the build environment to generate records linking each item to its parent in the product tree?

- [ ] **A** IUID
- [ ] **B** TR
- [ ] **C** LOT
- [ ] **D** SER

M1-04 Identity, link, and condition codes

The eight codes do three different jobs.

Identity: SER, LOT, IUID

These give a unit or a lot an identifier that the build environment records.

Link: TR

This links those identities into parent-child records up the product tree.

Condition: LL, SH, CSI, FC

These identify a condition of the item. They do not create records on their own.

Take two life-limited items. A landing gear shock strut has a landing cycle limit. Each strut counts its own cycles, so each strut needs its own serial number. It takes LL and SER.

An O-ring has a shelf-life limit. The limit attaches to the manufacturing batch, so the batch is what gets recorded. It takes LL and LOT.

Both items carry LL. LL says the item has a life limit. It does not say how the build environment will know which strut or which batch it holds. The identity code says that.

A condition code always needs an identity code beside it.

The requirement source decides which identity code. LL never decides it.

CHECK YOURSELF · M1-04

A designer applies LL to a new part and nothing else. What is missing?

- [ ] **A** TR, because every life-limited item needs parent-child records
- [ ] **B** CSI, because every life-limited part is a Critical Safety Item
- [ ] **C** An identity code, SER or LOT, set by the requirement source
- [ ] **D** Nothing. LL creates the life-tracking records on its own

M1-05 Which unit, or which lot

A serial number and a lot/batch number answer different questions.

Serial number

Which unit. One identifier belongs to one physical item. Use it when the thing that matters is per unit: remaining life, one specimen test result, a mandatory mark.

Lot/batch number

Which group. One identifier covers every item made from the same material, process run, or test sample. Use it when the property that matters is shared: a heat of metal, a cure batch, a shelf-life date, a source of supply.

EIA-649-1A DEFINITIONS

Unit identifier. "A sequentially issued identifier used to designate a specific instance of a product among like products. Examples: Item Unique Identification (IUID), serial number, Vehicle Identification Number (VIN)."

Group identifier. "Uniquely identifies a group of like units of the same product manufactured under uniform conditions. Examples: lot number/batch number."

Serial number. "A unique number identifying individual units within a series of like items. The serial number does not establish the PN or PIN but tracks the number of items that were produced under the PN or PIN. Serial numbers should be assigned to all items requiring tracking."

The standard also says when to use each one.

EIA-649-1A PRINCIPLES

Principle CI-5. “Individual units of a product are assigned a unique product unit identifier when there is a need to distinguish one unit of the product from another.”

Principle CI-7. “A series of like units of a product is assigned a unique product group identifier when it is unnecessary or impractical to identify individual units, but necessary to correlate units to a process, date, event, or test.”

Neither one substitutes for the other. A lot/batch number cannot say where one specific item went. A serial number cannot say which heat of material an item came from unless the lot is also recorded.

Some items need both. A premium selected material part needs a serial number to tie each test result to one part. It also needs a lot/batch record for heat traceability.

THE LOCATION GAP

A lot/batch number alone does not establish location. Fifty O-rings from one batch may go into fifty different assemblies. The lot record says what they share. It does not say where each one went.

That is why the **parent** of a lot/batch recorded item is serialized. The parent's serial number is the location. The lot record attaches to it.

CHECK YOURSELF · M1-05

Why is the parent of a Lot/Batch Recording Required item serialized?

- [ ] **A** The parent inherits the lot number and needs a serial number to carry it
- [ ] **B** A lot/batch number does not establish location, so the lot must attach to an identified unit
- [ ] **C** Serial numbers are cheaper for the build environment to record than lot numbers
- [ ] **D** Lot/batch numbers expire with the shelf life of the lot, and serial numbers never do

M1-06 Codes stack

Requirement sources and codes do not map one to one.

One source can drive several codes. Go back to the strut. Its landing cycle limit drives two codes. LL states the limit. SER gives each strut the serial number that tracks its own life.

Several sources can drive one code. A capacitor may carry a shelf-life limit. It may also fall under a counterfeit avoidance clause in the contract. Both sources call for lot/batch recording. One LOT code satisfies both. The reasons differ. The lot records are the same records.

Several sources can land on one item. An engine fire extinguisher cartridge has a shelf life. It is also explosive material, which needs special handling. Each source adds its own codes. Apply every code that each source

drives. Do not stop at the first source you find.

Many authorities, one parts list line

The sources come from authorities that do not coordinate with one another. The parts list line is where they meet. You will meet each authority later in the course.

Authority	Example sources	Taught in
FAA certification rules	Life limits. Airworthiness limitations. Structural and material rules.	Modules 3 and 4
DoD contract clauses	Item unique identification. Critical Safety Items. Counterfeit avoidance.	Modules 5, 6, and 7
Military standards	Fracture control. Identification marking. System safety.	Modules 6 and 7
Company engineering and safety	The cascade. Process and batch conditions. Special handling.	Modules 2 and 5

1. **Find every source**
   Ask what regulation, contract clause, designation, or physical condition applies to the item. There may be more than one.

2. **Apply every code each source drives**
   Do not duplicate a code. One LOT covers every source that needs lot/batch records.

3. **Apply the cascade**
   Once the item carries SER or LOT, every physical parent below the end item takes SER and TR.

CHECK YOURSELF · M1-06

An electronic part has a shelf-life limit and is bought under a contract with a counterfeit avoidance clause. How many LOT codes does it take?

- [ ] **A** None, because counterfeit avoidance requires serialization instead
- [ ] **B** Two, one for each requirement source
- [ ] **C** One LOT and one SER, because two sources need two identities
- [ ] **D** One, because both sources are satisfied by the same lot records

M1-07 Who does what, and who owns each source

Three organizations share this work. Each one owns a different piece.

Organization	Owns	Output
Design Engineering	Imposing the requirement	Part note codes and note texts on the parts list
Manufacturing and Production (the build environment)	Capturing the records	Serial numbers, lot/batch numbers, marks, and parent-child traceability records
Configuration Management	Reconciling the two	Confirmation that every code on the design has matching records from the build environment

Configuration Management does the reconciliation as part of **configuration status accounting**. The design is one side of the comparison. The build records are the other.

The part note code is the handoff between the three. If Design Engineering does not apply the code, the build environment has no instruction to capture the record. If the build environment does not capture the record, Configuration Management finds the gap at reconciliation. By then the program may already have delivered the unit.

A traceability requirement that never reaches the parts list never reaches the build environment.

FOR NON-ENGINEERS

You do not need to write part notes to use this course. Buyers flow the requirement to suppliers. Planners build it into work instructions. Inspectors verify the mark and the record. Technicians record the number at installation.

Every one of those actions starts from a part note code on a parts list. Knowing what the code means tells you what your step owes.

Who owns each requirement source

Take one main landing gear parts list. The shock strut's life limit comes from the Airworthiness Limitations Section. The outer cylinder's heat and lot requirement comes from the structural analysis. The brake hose's shelf life comes from the material specification. Three lines, three owners.

Design Engineering puts the codes on the parts list. It does not invent the requirements behind them. Other organizations own those requirements.

Source owner	What the owner supplies	Codes affected
Airworthiness and Certification	ALS entries, with the part, the limit, and the task. Too-small-to-mark findings.	SER, LL, LOT
Structures	Safe-life and damage-tolerant PSE lists. Fracture-critical part lists. Premium selection decisions.	SER, LOT, LL, FC
Materials and Processes	Castings, process and batch conditions, and shelf-life and cure-date limits. Coordination of explosive material.	LOT, LL, SH
Contracts	IUID, Critical Safety Item, and counterfeit avoidance clauses. The parent item level. The end item identity.	IUID, CSI, SER, LOT
Design control activity (customer)	Critical Safety Item designation	CSI
System Safety	Safety-Critical Item analysis. Inputs to Critical Safety Item designation.	None on the Safety-Critical Item basis
Radiation Health Protection	Identification of radioactive material, with the isotope and the amount	SH, LOT
Supply Chain	Flowdown of contract clauses to suppliers. Supplier lot, heat, and serial data.	LOT, SER, IUID
Design Engineering	The parts list. The final code set on each line.	All

When a code is missing, wrong, or in doubt, start with the owner of its source. Design Engineering corrects the parts list through change. Configuration Management confirms that the records match.

CHECK YOURSELF · M1-07

An engineer knows a bracket must be serialized but leaves the part note code off the parts list, planning to tell Manufacturing directly. What goes wrong?

- [ ] **A** The build environment has no design instruction, and reconciliation has nothing to check
- [ ] **B** Configuration Management adds the missing codes during reconciliation
- [ ] **C** Nothing, provided Manufacturing records the serial numbers anyway
- [ ] **D** Only a drafting error, which the checker corrects at the next drawing revision cycle

M1-08 Four questions set the codes

Go back to the strut line. Four questions produced its three codes.

1. **Which requirement sources apply?**
   The strut has a life limit in landing cycles. Its seals carry lot/batch numbers.

2. **Which identity code does each source drive?**
   A per-unit life needs a serial number. That is SER.

3. **Which condition codes apply?**
   A mandatory replacement limit adds LL.

4. **Where does the item sit in the tree?**
   The strut holds traceable children, so it takes TR. Its own parent takes SER and TR.

Ask the four questions for every line on a parts list. Each later module answers one or more of them.

Question	Answered in
1. Which requirement sources apply?	Modules 3 through 7
2. Which identity code does each source drive?	Modules 3 through 7
3. Which condition codes apply?	Module 3 for LL, Module 5 for SH, Module 7 for CSI and FC
4. Where does the item sit in the tree?	Module 2

Module 8 puts all four questions to work on a full parts list.

WHAT CONFIGURATION MANAGEMENT CHECKS

Every code on a parts list traces to a requirement source.

Every code carries its note text.

Later, Configuration Management compares these codes with the build records.

CHECK YOURSELF · M1-08

Which of the four questions decides whether an item takes TR?

- [ ] **A** Where does the item sit in the tree?
- [ ] **B** Which requirement sources apply?
- [ ] **C** Which condition codes apply?
- [ ] **D** Which identity code does each source drive?

M1-DR Parts-list drill

Read each line. Tick every part note code it takes. Then check it.

STANDING ASSUMPTIONS FOR EVERY LINE

Unless the line says otherwise, the item has no children that carry SER or LOT, it is not the end item, and the contract carries no IUID clause.

A line is right only when the set is exact. A missing code is a defect. An extra code is also a defect. Tick **No code** when nothing applies.

LINE M1-L1 · FIND 12 · PART NO. EX-4020-PLT

PLATE, ALUMINUM ALLOY (RAW MATERIAL)

- A raw material callout on a detail part's parts list.
- The material specification requires heat and lot traceability.

☐ SER ☐ LOT ☐ TR ☐ IUID ☐ LL ☐ SH ☐ CSI ☐ FC ☐ No code

LINE M1-L2 · FIND 1 · PART NO. EX-5512-300

STRUT, SHOCK, MAIN LANDING GEAR

- Life-limited by landing cycles.
- Its parts list includes lot-recorded seals.

☐ SER ☐ LOT ☐ TR ☐ IUID ☐ LL ☐ SH ☐ CSI ☐ FC ☐ No code

LINE M1-L3 · FIND 9 · PART NO. EX-2201-041

COVER, ACCESS PANEL

- No requirement source applies.
- None of its children carry SER or LOT.

☐ SER ☐ LOT ☐ TR ☐ IUID ☐ LL ☐ SH ☐ CSI ☐ FC ☐ No code

LINE M1-L4 · FIND 14 · PART NO. EX-8840-007

CANDLE, EMERGENCY OXYGEN

- Shelf-life limit attached to the manufacturing batch.
- No certification or contract requirement calls for serialization.

☐ SER ☐ LOT ☐ TR ☐ IUID ☐ LL ☐ SH ☐ CSI ☐ FC ☐ No code

M1-EX Exercises

Write your reasoning, not just your conclusion. The reasoning is what is being assessed, and several of these have more than one defensible answer. Where a question turns on a fact you have not been given, **say so, and say what you would ask for and whom you would ask.** Recognizing the missing fact is part of the exercise.

Exercise 1: The case for the records

A program manager proposes to drop lot/batch recording on hydraulic O-rings to save cost. The argument: O-rings are cheap, and they are replaced at every overhaul anyway. The hydraulic assemblies would stay serialized.

a. Describe a concrete failure that lot/batch records on the O-rings would let the program contain. Then explain what happens to the containment if the records are gone.

b. The manager says the serialized assemblies are enough. What do the assembly serial numbers give you on their own, and what do they lose without the O-ring lot records?

c. Explain the role of the assembly's serial number in making the O-ring lot records useful.

d. Who owns the decision to remove the code from the parts list? If the code were dropped without that authority, who would find the gap, and when?

MODULE 2

The cascade to the end item

How records run from each traceable item up every physical parent to the end item

M2-01 One missing parent

WHAT YOU WILL BE ABLE TO DO

Apply SER and TR to every physical parent below the end item.

Code an installation drawing and an end item.

Choose the long or short note text for SER, LOT, and TR.

Explain how one uncoded parent breaks the chain of records.

A brake hose on a main landing gear has a shelf life. The hose is lot/batch recorded. Its parts list line is right.

The hose goes into a brake line assembly. The brake line assembly has no requirement source of its own. Its parts list line carries no code.

Two years after delivery, the hose supplier reports a cure defect in one lot. The fleet team asks a simple question. Which airplanes have hoses from that lot?

The records cannot answer. They show which lot reached the factory. They stop at a brake line assembly with no serial number. Every airplane in the effectivity range now needs an inspection.

Nobody made an error about the hose. The gap was one level up, on a part that looked like it needed nothing.

What each record answers

Fix the gap, and the chain of records works. Each record answers one question.

- The **lot number** says which group the hose came from: the supplier, the batch, and the cure date.
- The **brake line assembly serial number** says which unit holds the hose.
- The **main landing gear serial number** says which gear holds that unit.
- The **tail number** says which airplane carries the gear.

A fleet team does not ask “which lot?” It asks “which airplanes?” Only the full chain answers that question.

Cost rises at each stage

When the gap surfaces	What happens	Typical cost
Design release	The checker adds the codes before release.	Minutes
Production planning	Design Engineering revises the parts list through change. Planning updates the work instructions.	Hours to days
After build	Someone reconstructs the records. Technicians read serial numbers off installed parts, sometimes with panels removed.	Days to weeks per airplane
Delivery and audit	An audit or a customer inspection finds the gap. The customer may hold delivery.	Weeks, plus schedule exposure
In service	A lot or serial question arrives from the field. The fleet team searches or inspects every airplane in the effectivity range.	Fleet action

These costs show the shape of the curve. They are illustrative, not measured.

In the brake hose case, the fix at release was two codes on one line.

CHECK YOURSELF · M2-01

Put the records in the order a fleet team follows them, from the suspect hose to the airplane.

Number the items in order, 1 to 4 (1 = Start: the suspect hose).

____ Tail number

____ Hose lot number

____ Main landing gear serial number

____ Brake line assembly serial number

M2-02 The governing rule

One rule governs every item that carries SER or LOT.

When an item holds a serialization recording requirement or a lot/batch recording requirement, serialization and traceability records must be maintained up the product tree to the end item.

This is a company directive. It applies whatever the requirement source. There is no threshold and no shorter scope.

We call this rule the **cascade**. The records cascade up the product tree, one level at a time, to the end item.

The rule has three consequences.

- **Every physical parent is serialized.** The item's next-higher physical parent is serialized. So is every higher physical parent below the end item.
- **Every physical parent generates traceability records.** Each record links an item to its next-higher physical parent.
- **The chain of records stops at the end item.** The end item is the top-level asset or system under contract. It is typically the complete aircraft, by tail number, or the missile system.

Only SER and TR cascade. Every other code stays on the item that holds it. A condition code such as LL describes the item it sits on, so a parent never inherits it.

The end item is not a Line Replaceable Unit (LRU). Mechanics remove and replace an LRU on the flight line. The chain of records does not stop there. It continues to the aircraft or system that holds the LRU.

Later pages in this module walk the cascade one level at a time. For now, hold the rule.

CHECK YOURSELF · M2-02

A serialized valve goes into a manifold. The manifold goes into a Line Replaceable Unit, and the LRU goes into an aircraft. Where do the serialization and traceability records stop?

- [ ] **A** At the LRU, since it is the replaceable unit
- [ ] **B** At the aircraft, the end item under contract
- [ ] **C** At the manifold, the valve's next-higher parent
- [ ] **D** At the valve, since only the valve holds its own requirement

M2-03 What the TR code directs

SER and LOT give an item an identity. TR puts that identity somewhere.

The Traceability Records Required code, TR, directs the build environment to generate part traceability records. These records document where each Serialization Recording Required item and each Lot/Batch Recording Required item is installed.

Each traceability record links an item to its next-higher physical parent. String the records together, and they answer the question behind the whole system: this part, in which aircraft?

A link has to identify both ends. On one end, the child's part number with its serial number or lot/batch number. On the other end, the physical parent's part number and serial number.

One link is not enough. A record that says “valve serial 0412 is in manifold serial 0031” is useless if nobody knows where manifold 0031 went. The next link says that. So does the one after it, all the way to the end item.

TR BELONGS ON THE PARENT

TR goes on the item whose parts list holds traceable children. It tells the build environment to record which child went into this unit.

An item with no traceable children has nothing to link below it. It does not take TR, even if it carries SER itself.

CHECK YOURSELF · M2-03

A serialized relief valve has no children. Does the valve itself take TR?

- [ ] **A** Yes, because the long SER note requires traceability
- [ ] **B** Yes. Every serialized item takes TR
- [ ] **C** No. TR goes on the parent whose children carry SER or LOT
- [ ] **D** No, because a valve with no children is not a traceable item at all

M2-04 Every physical parent

Recall the governing rule. When an item holds a serialization recording requirement or a lot/batch recording requirement, serialization and traceability records must be maintained up the product tree to the end item.

Here is what that means for the parents. A serialized item's next-higher physical parent must also be serialized. So must every higher physical parent below the end item. The parent of a lot/batch recorded item is serialized too, because a lot/batch number alone does not establish location.

For each physical parent below the end item, apply:

SER SERIALIZATION RECORDING REQUIRED.

TR TRACEABILITY RECORDS REQUIRED.

The rule does not ask how critical the child is, how cheap the parent is, or how deep the tree runs. One traceable item at the bottom serializes every physical parent above it.

Notice the words **physical** and **below the end item**. Two items in a product tree do not take SER and TR from the cascade. An installation drawing is not a physical part: it is not consumed into the product. The end item is where the chain of records stops. The next section covers both.

CHECK YOURSELF · M2-04

A lot-recorded O-ring sits five levels below the aircraft. None of the four assemblies between them holds a requirement of its own. How many of those assemblies take SER and TR?

- [ ] **A** None, because a lot-recorded child does not drive serialization of its parent
- [ ] **B** Only the assemblies that are Line Replaceable Units
- [ ] **C** All four physical parents below the end item
- [ ] **D** Only the O-ring's next-higher physical parent

M2-05 Long and short note texts

SER and LOT each have a long note text and a short one. The choice depends on why the item carries the code.

An item that holds **its own** serialization recording requirement takes the long SER text:

SER `SERIALIZATION RECORDING REQUIRED. THIS ITEM IS SERIALIZED AND TRACEABILITY MUST BE MAINTAINED BETWEEN THIS ITEM AND ALL HIGHER-LEVEL PHYSICAL ITEMS UP TO THE END ITEM.`

If that item also has children that carry SER or LOT, it also takes:

TR `TRACEABILITY RECORDS REQUIRED.`

A physical parent that carries SER **only because of the cascade** takes the short texts:

SER `SERIALIZATION RECORDING REQUIRED.`

TR `TRACEABILITY RECORDS REQUIRED.`

LOT follows the same logic. A raw material callout, such as bar stock, sheet, or plate, takes the short text:

LOT `LOT/BATCH RECORDING REQUIRED.`

A detail part or a vendor part that holds a lot/batch recording requirement takes the long text:

LOT `LOT/BATCH RECORDING REQUIRED. THIS ITEM REQUIRES LOT/BATCH TRACEABILITY AND TRACEABILITY MUST BE MAINTAINED BETWEEN THIS ITEM AND ALL HIGHER-LEVEL PHYSICAL ITEMS UP TO THE END ITEM.`

The long texts carry the cascade inside the note. They tell the build environment that traceability runs from this item to the end item.

CHECK YOURSELF · M2-05

An assembly has no traceability requirement of its own. One of its children is a serialized life-limited actuator. Which codes does the assembly take?

- [ ] **A** TR only, since the assembly holds no requirement of its own
- [ ] **B** LL and SER, inherited from the actuator
- [ ] **C** SER with the long note text, and no TR
- [ ] **D** SER and TR, with the short note texts

M2-06 Installation drawings and the end item

Two items in the product tree break the pattern.

Installation drawings. An installation drawing depicts installation instructions. It is not consumed into the production system as a physical part. You cannot serialize an instruction. For each installation drawing that sits between a traceable item and the end item, apply:

TR `TRACEABILITY RECORDS REQUIRED. THIS INSTALLATION DRAWING DEPICTS TRACEABLE PARTS WHOSE SERIAL NUMBERS MUST BE TRACED TO ALL HIGHER-LEVEL PHYSICAL ITEMS UP TO THE END ITEM.`

The installation drawing takes TR and no SER. The chain of records passes through it and reaches the next physical item or the end item.

The end item. The end item is the top-level asset or system under contract. It is typically the complete aircraft, by tail number, or the missile system. For the end item, apply:

TR `TRACEABILITY BEYOND THIS LEVEL IS NOT REQUIRED.`

The end item never takes SER. The contract assigns its identity, such as a tail number or a system serial number.

Some contracts require IUID for the end item. Then the end item also takes the IUID code. It still takes no SER. Module 6 covers IUID. Its last page shows which IUID note text the end item takes.

Item in the tree	Codes	Note text
Physical parent below the end item	SER, TR	Short SER and short TR
Installation drawing between a traceable item and the end item	TR	Installation TR text
End item, no IUID requirement	TR	Terminator TR text
End item, contract requires IUID	TR, IUID	Terminator TR text and an IUID note text. Never SER.

CHECK YOURSELF · M2-06

A flap installation drawing sits between a serialized flap drive unit and the aircraft. Which codes does the installation drawing take?

- [ ] **A** TR, with the installation note text
- [ ] **B** SER and TR, like any other parent
- [ ] **C** TR, with the terminator note text
- [ ] **D** No code, because installation drawings are not parts

M2-07 A worked cascade

Walk one tree from the bottom up. An actuator sits inside a flap drive unit. The flap drive unit is installed on the aircraft through a flap installation drawing.

Aircraft, tail number assigned by contract — TR

End item. Terminator TR text. It never takes SER. If the contract required IUID for it, it would add IUID.

└ **Flap installation** — TR

Installation drawing. Installation TR text.

└ **Flap drive unit** — SER TR

A Line Replaceable Unit. Physical parent. Short texts. The chain of records does not stop here.

└ **Actuator assembly** — SER TR

Physical parent of traceable children. Short texts.

└ **Housing, Fracture-Critical Traceable Part** — FC SER LOT

Holds its own requirements. Long SER and long LOT texts.

└ **O-ring set, shelf-life limited** — LL LOT

Age and shelf-life LL text. Long LOT text.

└ **Fasteners, heat and lot traceability** — LOT

Long LOT text. No LL: fasteners have no shelf life.

└ **End cap** — no code

No source applies. No code.

1. **Code the children from their own sources**
   The housing, the O-rings, and the fasteners each carry codes from their own requirement sources. The end cap carries none.

2. **Go up one level**
   The actuator has traceable children. It takes SER and TR.

3. **Keep going up through every physical parent**
   The flap drive unit is a physical parent of a serialized actuator. SER and TR. Being an LRU changes nothing.

4. **Pass through the installation drawing**
   TR with the installation text. No SER.

5. **Stop at the end item**
   TR with the terminator text. The contract gives the aircraft its identity.

Notice what did not cascade. FC stayed on the housing. LL stayed on the O-rings. Condition codes describe the item they sit on. Only SER and TR cascade.

If you keep one step from this module, keep the parent check. For every SER or LOT line, look one level up. That physical parent needs SER and TR.

WHAT CONFIGURATION MANAGEMENT CHECKS

Every SER or LOT line has a physical parent above it.

Each physical parent carries SER and TR.

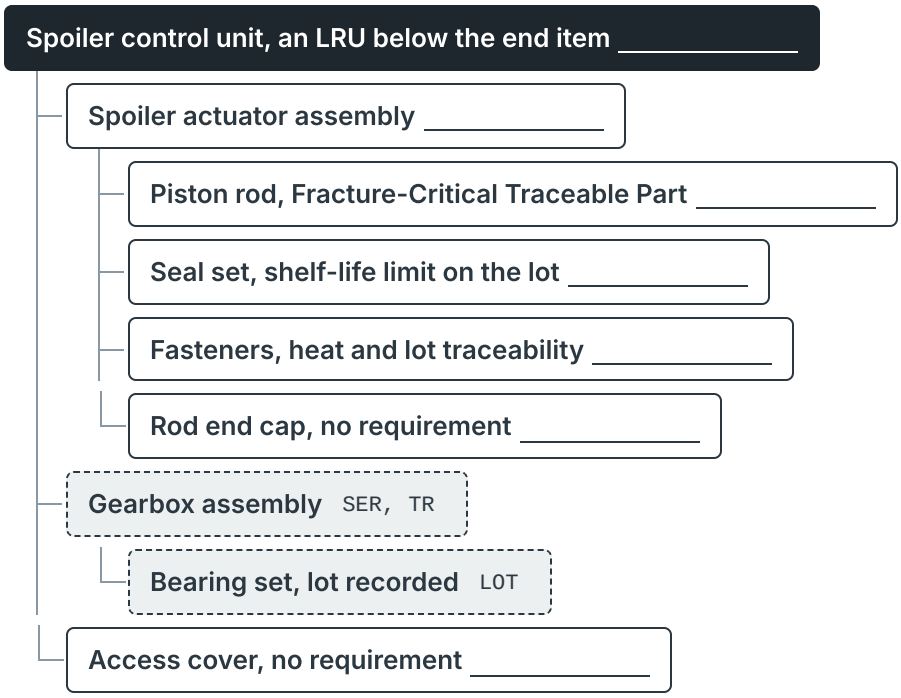
Each installation drawing carries the installation TR text.

The chain of records reaches the end item without a break.

CHECK YOURSELF · M2-07

Here is a second tree. A spoiler control unit sits below the end item. The piston rod's standard requires a serial number and a material lot. Label each block with its codes. Two blocks are already labeled.

Write the right label in each box. Labels: **SER, TR · FC, SER, LOT · LL, LOT · LOT · No code · TR (installation text) · TR (terminator text)**



M2-DR Parts-list drill

Read each line. Tick every part note code it takes. Then check it.

STANDING ASSUMPTIONS FOR EVERY LINE

Unless the line says otherwise, the item has no children that carry SER or LOT, it is not the end item, and the contract carries no IUID clause.

A line is right only when the set is exact. A missing code is a defect. An extra code is also a defect. Tick **No code** when nothing applies.

LINE M2-L1 · FIND 3 · PART NO. EX-7720-114

MANIFOLD ASSY, HYDRAULIC

- No requirement source applies to the manifold itself.
- Its children include a relief valve that carries SER.

☐ SER ☐ LOT ☐ TR ☐ IUID ☐ LL ☐ SH ☐ CSI ☐ FC ☐ No code

LINE M2-L2 · FIND 6 · PART NO. EX-3308-020

HARNESS ASSY, WIRING

- No requirement source applies to the harness itself.
- Its children include connectors that carry LOT.

☐ SER ☐ LOT ☐ TR ☐ IUID ☐ LL ☐ SH ☐ CSI ☐ FC ☐ No code

LINE M2-L3 · FIND 2 · PART NO. EX-5500-100

UNIT, FLAP DRIVE (LRU)

- A Line Replaceable Unit.
- Its children include a serialized actuator.
- No requirement of its own.

☐ SER ☐ LOT ☐ TR ☐ IUID ☐ LL ☐ SH ☐ CSI ☐ FC ☐ No code

LINE M2-L4 · FIND 1 · PART NO. EX-IN-5500

INSTALLATION, FLAP

- An installation drawing between a serialized flap drive unit and the end item.

☐ SER ☐ LOT ☐ TR ☐ IUID ☐ LL ☐ SH ☐ CSI ☐ FC ☐ No code

LINE M2-L5 · FIND N/A · PART NO. EX-TAIL-ASSIGNED

AIRPLANE, COMPLETE (END ITEM)

- The top-level asset under contract.
- The contract carries no IUID clause.
- Traceable items are installed beneath it.

☐ SER ☐ LOT ☐ TR ☐ IUID ☐ LL ☐ SH ☐ CSI ☐ FC ☐ No code

M2-EX Exercises

Write your reasoning, not just your conclusion. The reasoning is what is being assessed, and several of these have more than one defensible answer. Where a question turns on a fact you have not been given, **say so, and say what you would ask for and whom you would ask.** Recognizing the missing fact is part of the exercise.

Exercise 1: Reading a parts list

An actuator assembly parts list shows five items. A housing that is a Fracture-Critical Traceable Part; its governing standard requires each housing to be serialized and its material to be lot/batch recorded. An end cap with no requirement. A set of O-rings with a shelf-life limit. An aluminum bar stock callout that requires lot traceability. A set of fasteners that require heat and lot traceability. The actuator goes into a flap drive unit, the flap drive unit goes into the wing, and the wing goes into the aircraft.

a. For the housing and the O-rings, name each code and say whether it is an identity code or a condition code.

b. Which items take the long LOT note text and which take the short one? Say why.

c. Which codes does the actuator assembly itself take, and in which note form?

d. How far do the records cascade? Which codes do the flap drive unit and the wing take?

MODULE 3

Life limits

The LL code, the 14 CFR 43.10 definitions, cycle and shelf-life limits, and control of removed life-limited parts

M3-01 The LL code and 14 CFR 43.10

WHAT YOU WILL BE ABLE TO DO

Decide whether a limit makes a part life-limited under 14 CFR 43.10.

Pick the identity code a life limit needs, SER or LOT.

Choose the cycle/usage or age/shelf-life LL note text.

Name the ways to control a removed life-limited part.

A turbine disk must come out after 20,000 engine cycles. An O-ring must not go in after its shelf life runs out. Both carry LL. The records track only one of them by serial number.

The Life Limited code, LL, identifies the item as life-limited and states its limit. It does not create records on its own. Apply it with the identity code that the requirement source drives.

14 CFR 43.10 supplies the definitions.

14 CFR 43.10(A)

Life-limited part means “any part for which a mandatory replacement limit is specified in the type design, the Instructions for Continued Airworthiness, or the maintenance manual.”

Life status means “the accumulated cycles, hours, or any other mandatory replacement limit of a life-limited part.”

The limit may be expressed in cycles, hours, or calendar time. The form of the limit sets which LL note text you use. It does not set the identity code.

There are two categories of life-limited item.

Cycle/usage-controlled

Limited by cycles or operating hours. Remaining life is a per-unit quantity. Serialized.

Age/shelf-life-controlled

Limited by shelf life, cure date, or storage age. The limit usually attaches to the manufacturing lot/batch. Lot/batch recorded, unless another source requires a serial number.

LL states the limit. The requirement source sets the identity code.

CHECK YOURSELF · M3-01

Where can a mandatory replacement limit be specified and still make a part life-limited under 14 CFR 43.10?

- [ ] **A** Only in the type certificate data sheet issued by the FAA for the product
- [ ] **B** In the type design, the Instructions for Continued Airworthiness, or the maintenance manual
- [ ] **C** Anywhere the manufacturer recommends a replacement in its service literature
- [ ] **D** Only in the Airworthiness Limitations Section

M3-02 Cycle/usage-controlled items

A Cycle/Usage-Controlled Life-Limited Item is an item life-limited by cycles or operating hours. It is serialized, because remaining life is a per-unit quantity. Two identical disks on the same shelf can have very different remaining lives.

Common examples:

Landing gear

Landing gear assemblies. Main gear shock struts and oleo cylinders. Main wheel axles. Side stays and drag braces.

Engines and power

Turbine disks. Compressor disks. Turbine blades. Auxiliary power unit components.

Systems and actuation

Flap and slat actuator ball screws. Air cycle machine turbines and compressors. Ram air turbine deploy actuators. Primary bleed air valves. Pitch change actuator piston assemblies.

For a Cycle/Usage-Controlled Life-Limited Item, apply:

LL LIFE LIMITED PART. THIS PART IS CYCLE/USAGE-CONTROLLED AND IS LIMITED TO A SPECIFIC NUMBER OF USAGES. [NUMBER OF FLIGHTS, CYCLE LIMIT, ETC.].

SER SERIALIZATION RECORDING REQUIRED. THIS ITEM IS SERIALIZED AND TRACEABILITY MUST BE MAINTAINED BETWEEN THIS ITEM AND ALL HIGHER-LEVEL PHYSICAL ITEMS UP TO THE END ITEM.

The bracketed field states the limit. A number of flights, a cycle limit, or operating hours. Write the number, not a pointer to a document that someone must go find.

CHECK YOURSELF · M3-02

A compressor disk is limited to 20,000 engine cycles. Which codes apply?

- [ ] **A** LL with the age and shelf-life note text, and LOT
- [ ] **B** SER only, because the disk is tracked by engine
- [ ] **C** LL only, because the engine log tracks the cycles
- [ ] **D** LL with the cycle/usage note text, and SER

M3-03 Age/shelf-life-controlled items

An Age/Shelf-Life-Controlled Life-Limited Item is an item life-limited by shelf life, cure date, or storage age. It is lot/batch recorded, because the limit attaches to the manufacturing lot/batch rather than to the individual unit.

Common examples:

Seals and hoses

O-rings. Gaskets. Rubber-based hoses, such as hydraulic hoses and fuel hoses.

Electrical

Electronic components (EEE). Microchips. Capacitors. Diodes. Lithium batteries. Standard circuit breakers. Relays.

Chemicals and composites

Epoxy resins used in structural bonding. Prepregs. Polysulfide or polythioether gap and fuel tank sealants. Polyurethane topcoats. Chromate and non-chromate primers. Synthetic grease for actuator screws, landing gear pivots, or bearings.

Emergency and survival

Emergency oxygen candles. Fire extinguisher squibs. Oxygen mask rebreather bags. Medical kits. Life vests and life jackets.

Filters

Filters.

Not on this list

Specialized metallic fasteners. They have no shelf life. They carry heat and lot traceability under process and batch conditions.

A shelf life, cure date, or storage age limit that attaches to the manufacturing lot/batch. Use this pairing when no certification or contract requirement calls for serialization. Apply:

LL LIFE LIMITED PART. THIS PART IS AGE/SHELF-LIFE-CONTROLLED IRRESPECTIVE OF HOW MANY TIMES IT IS USED. [SHELF LIFE, CURE DATE LIMIT, OR STORAGE AGE LIMIT].

LOT LOT/BATCH RECORDING REQUIRED. THIS ITEM REQUIRES LOT/BATCH TRACEABILITY AND TRACEABILITY MUST BE MAINTAINED BETWEEN THIS ITEM AND ALL HIGHER-LEVEL PHYSICAL ITEMS UP TO THE END ITEM.

A calendar limit does not by itself make an item a lot/batch item. Module 4 shows the exception. A discrete part with a calendar limit in the Airworthiness Limitations Section keeps its serial number. It takes LL with the age and shelf-life note text, and SER.

CHECK YOURSELF · M3-03

Oxygen mask rebreather bags carry a shelf-life limit on the manufacturing lot. Nothing else calls for serialization. Which codes apply?

- [ ] **A** LL with the age and shelf-life note text, and SER
- [ ] **B** LL with the cycle/usage note text, and SER
- [ ] **C** LOT only
- [ ] **D** LL with the age and shelf-life note text, and LOT

M3-04 Controlling a removed life-limited part

14 CFR 43.10(c) requires each person who removes a life-limited part from a type-certificated product to control it. The method must deter installation after the part reaches its life limit. The acceptable methods are:

Method	What it takes
Record keeping system	Part number, serial number, and current life status, updated at each removal
Tag or record attached to the part	Part number, serial number, and current life status
Non-permanent marking	Current life status, marked legibly, updated at each removal
Permanent marking	Current life status, permanently marked
Segregation	Physical separation from eligible parts, with records
`Mutilation`	Rendering the part beyond repair and incapable of appearing airworthy
Other method	Any method approved or accepted by the FAA

When the part is later sold or transferred, the mark, tag, or record goes with it, unless the part is mutilated first.

Most of these methods need a part number and a serial number. Serialization makes 43.10 compliance workable. The serialization requirement itself comes from 14 CFR 45.15(c), when the limit is in the Airworthiness Limitations Section.

BACKGROUND · NOT TESTED · 14 CFR 45.16

The marking methods rely on 14 CFR 45.16. When a person who must comply with 43.10 asks, the holder of the type certificate or design approval for a life-limited part must respond. It must provide marking instructions. Or it must state that the part cannot be practicably marked without compromising its integrity. That is a design responsibility.

SHELF LIFE IS A DIFFERENT REGIME

A shelf life, cure date, or storage age limit on material held in storage is not, by itself, a 14 CFR 43.10 mandatory replacement limit.

The LL code still applies to age/shelf-life-controlled items. The code covers both categories. The regulation does not. Know which regime a limit sits in before you cite 43.10.

WHAT CONFIGURATION MANAGEMENT CHECKS

Every LL line pairs with SER or LOT.

Every LL note text states the limit as a number.

Each life-limited item has the starting data its life status needs, such as a cure date or the cycles at installation.

CHECK YOURSELF · M3-04

Why does serialization make 14 CFR 43.10 control of removed life-limited parts workable?

- [ ] **A** The FAA assigns a serial number to each life-limited part when it is removed
- [ ] **B** 14 CFR 43.10 itself requires every life-limited part to carry a serial number
- [ ] **C** Most of the acceptable control methods need a part number and a serial number
- [ ] **D** A serial number lets the part be sold or transferred without its tag or record

M3-DR Parts-list drill

Read each line. Tick every part note code it takes. Then check it.

STANDING ASSUMPTIONS FOR EVERY LINE

Unless the line says otherwise, the item has no children that carry SER or LOT, it is not the end item, and the contract carries no IUID clause.

A line is right only when the set is exact. A missing code is a defect. An extra code is also a defect. Tick **No code** when nothing applies.

LINE M3-L1 · FIND 3 · PART NO. EX-HPT-1ST-DSK

DISK, TURBINE, STAGE 1

- Life-limited to 15,000 engine cycles.
- No other source applies.

☐ SER ☐ LOT ☐ TR ☐ IUID ☐ LL ☐ SH ☐ CSI ☐ FC ☐ No code

LINE M3-L2 · FIND 6 · PART NO. EX-AXL-MW-30

AXLE, MAIN WHEEL

- Life-limited to 25,000 landings.

☐ SER ☐ LOT ☐ TR ☐ IUID ☐ LL ☐ SH ☐ CSI ☐ FC ☐ No code

LINE M3-L3 · FIND 17 · PART NO. EX-RBR-BAG-4

BAG, REBREATHER, OXYGEN MASK

- Shelf-life limit on the manufacturing lot.
- No certification or contract requirement calls for serialization.

☐ SER ☐ LOT ☐ TR ☐ IUID ☐ LL ☐ SH ☐ CSI ☐ FC ☐ No code

LINE M3-L4 · FIND 11 · PART NO. EX-HOS-HYD-08

HOSE ASSY, HYDRAULIC, RUBBER

- Shelf-life limit on the manufacturing lot.
- No certification or contract requirement calls for serialization.

☐ SER ☐ LOT ☐ TR ☐ IUID ☐ LL ☐ SH ☐ CSI ☐ FC ☐ No code

LINE M3-L5 · FIND 2 · PART NO. EX-BKT-VLV-14

BRACKET ASSY, VALVE MOUNTING

- No requirement source applies to the bracket assembly itself.
- Its children include a cycle-limited valve that carries SER.

☐ SER ☐ LOT ☐ TR ☐ IUID ☐ LL ☐ SH ☐ CSI ☐ FC ☐ No code

M3-EX Exercises

Write your reasoning, not just your conclusion. The reasoning is what is being assessed, and several of these have more than one defensible answer. Where a question turns on a fact you have not been given, **say so, and say what you would ask for and whom you would ask.** Recognizing the missing fact is part of the exercise.

Exercise 1: The removed actuator

A mechanic removes a flap actuator from an airplane for repair. The actuator is a Cycle/Usage-Controlled Life-Limited Item. It has 2,000 cycles of life left. The mechanic sets it on a shelf. A second actuator with the same part number sits on the same shelf. It has 300 cycles left.

a. Which codes does the actuator's parts list line take? Explain the identity code you chose.

b. Name three ways the shop could control the removed actuator. Which of them need a serial number?

c. The two actuators look identical. What goes wrong if neither carries a serial number? Who finds out, and when?

MODULE 4

Serialization sources

Where serialization requirements come from: Airworthiness Limitation Items, Principal Structural Elements, and premium selected materials, and how each one sets the codes

M4-01 Where serialization comes from

WHAT YOU WILL BE ABLE TO DO

Code any ALS entry with the three questions.

Code safe-life and damage-tolerant PSEs and their materials.

Explain why a premium selected part needs both SER and LOT.

A landing gear axle, an oxygen cylinder, and a forged fitting all take SER. The axle wears out by landings. The cylinder carries an entry in the Airworthiness Limitations Section. The fitting earned a higher design value from its own test. Three reasons, one code.

SER drives one action: record serial numbers, or apply and record serial numbers. The requirement behind that action comes from many places.

Source	Why it needs a serial number
Cycle/usage-controlled life-limited items	Remaining life is a per-unit quantity
Airworthiness Limitation Items on a discrete part	14 CFR 45.15(c) requires the serial number
Safe-life Principal Structural Elements, and discrete damage-tolerant PSE parts	An ALS entry attaches to a discrete part
Premium selected materials	Each test result must tie to one part instance
Critical Safety Items	Serialization reaches them through IUID
Fracture-Critical Traceable Parts	MIL-STD-1530 names serialization as a control
DoD serially managed items and parent items	Every UII carries a serial number
Every physical parent of a serialized or lot/batch recorded item	The cascade to the end item

The sources differ. The result does not. Once an item carries SER, the cascade to the end item applies.

This module covers sources from two parts of 14 CFR. Part 25 holds the airworthiness standards for transport category airplanes. Part 45 holds the marking rules.

CHECK YOURSELF · M4-01

Which of these does NOT, on its own, drive serialization of the item?

- [ ] **A** An Airworthiness Limitations Section entry on a discrete part
- [ ] **B** A premium selection of material under 14 CFR 25.613(e)
- [ ] **C** A shelf-life limit that attaches to the manufacturing lot
- [ ] **D** Being the physical parent of a lot/batch recorded item

M4-02 Airworthiness Limitation Items

Start with where the limit is published.

14 CFR 25.1529 requires Instructions for Continued Airworthiness (ICA), prepared in accordance with Appendix H to Part 25. Paragraph H25.4 of Appendix H establishes the content of the Airworthiness Limitations Section (ALS). The limits in it are mandatory.

BACKGROUND · NOT TESTED

Appendix H also requires a statement in a prominent location in the ALS. The statement says the ALS is FAA-approved. It also says the ALS specifies maintenance required under 14 CFR 43.16 and 91.403, unless the FAA has approved an alternative program. That statement makes the limits mandatory for operators.

DEFINITION

An **Airworthiness Limitation Item (ALI)** is an item holding a mandatory replacement time, structural inspection interval, or related procedure approved as part of the type design and published in the Airworthiness Limitations Section (ALS) of the Instructions for Continued Airworthiness (ICA).

The marking rule is in 14 CFR 45.15(c), under the heading Critical parts:

14 CFR 45.15(C)

“Each person who manufactures a part for which a replacement time, inspection interval, or related procedure is specified in the Airworthiness Limitations section of a manufacturer’s maintenance manual or Instructions for Continued Airworthiness must permanently and legibly mark that part with a serial number (or equivalent) unique to that part in addition to the other applicable requirements of this section.”

Read the rule for its triggers. There are three. A replacement time. An inspection interval. A related procedure. Any one of them requires the serial number.

The trigger is the ALS entry. The form of the limit does not matter.

CHECK YOURSELF · M4-02

An ALS entry gives a discrete fitting a structural inspection interval. It carries no replacement time. Does 14 CFR 45.15(c) require a serial number on the fitting?

- [ ] **A** No. Structural inspections are always tracked by airplane, not by part
- [ ] **B** No. Only a replacement time triggers the rule
- [ ] **C** Only if the fitting is also a Critical Safety Item
- [ ] **D** Yes. An inspection interval is one of the three triggers

M4-03 Three questions set the codes

Three questions set the codes for an ALI. They are independent. Answer all three.

1. **What does the ALS entry attach to?**
   This sets the identity code. A discrete part takes SER. An airframe location or zone with no separately manufactured part takes no code.

2. **Does the entry carry a mandatory replacement limit?**
   This sets whether LL applies, and nothing else. A replacement limit adds LL. An inspection interval or a related procedure alone does not.

3. **How is the replacement limit expressed?**
   This sets which LL note text to use. It does not change the identity code. Answer it only when the second answer is yes.

The ALS entry attaches to	The entry carries	Codes	LL note text
A discrete part	A mandatory replacement limit in cycles, landings, or flight hours	SER, LL	Cycle/usage
A discrete part	A mandatory replacement limit in calendar time	SER, LL	Age and shelf-life
A discrete part	An inspection interval or a related procedure only	SER	None
An airframe location or zone, with no separately manufactured part	Any of the three triggers	None	None
A manufactured lot/batch of material, not a discrete part	A mandatory replacement limit (uncommon)	LOT, LL	Per the limit expressed

A part often carries both a replacement limit and an inspection interval. Inspect every 5,000 cycles, replace at 30,000 cycles, for example. Use the row for the replacement limit. The inspection interval adds no code of its own.

An ALI part may also be a Critical Safety Item, a Fracture-Critical Traceable Part, or a premium selected material part. The codes stack. Apply every code each source drives.

CHECK YOURSELF · M4-03

An ALS entry for a discrete lug reads: inspect every 5,000 flight cycles, replace at 30,000 flight cycles. Which codes apply?

- [ ] **A** SER and LL, with the cycle/usage LL note text
- [ ] **B** LOT and LL, because the limit is stated in cycles
- [ ] **C** SER and two LL notes, one for each limit
- [ ] **D** SER only, because the inspection interval comes first

M4-04 Calendar limits and parts too small to mark

The third question is the one most often misread.

An oxygen cylinder carries an ALS replacement limit of 5 years. The limit is calendar time. Calendar time sounds like shelf life, and shelf life sounds like lot/batch recording. That line of reasoning is wrong.

A calendar-limited ALI on a discrete part is still serialized.

14 CFR 45.15(c) does not care how the limit is expressed. Calendar time changes the LL note text. It does not move the part to lot/batch recording.

The oxygen cylinder takes SER and LL, with the age and shelf-life LL note text:

LL LIFE LIMITED PART. THIS PART IS AGE/SHELF-LIFE-CONTROLLED IRRESPECTIVE OF HOW MANY TIMES IT IS USED. [SHELF LIFE, CURE DATE LIMIT, OR STORAGE AGE LIMIT].

SER SERIALIZATION RECORDING REQUIRED. THIS ITEM IS SERIALIZED AND TRACEABILITY MUST BE MAINTAINED BETWEEN THIS ITEM AND ALL HIGHER-LEVEL PHYSICAL ITEMS UP TO THE END ITEM.

Some ALI parts are too small to mark. 14 CFR 45.15(d) covers them:

14 CFR 45.15(D)

“If the FAA finds a part or article is too small or otherwise impractical to mark with any of the information required by this part, the manufacturer must attach that information to the part or its container.”

The part still carries a unique serial number. The number moves to a tag or to the container. Lot/batch recording does not replace it.

CHECK YOURSELF · M4-04

A small ALI spring has a calendar replacement limit and is too small to mark. A planner proposes lot/batch recording instead of serialization. What is the correct response?

- [ ] **A** Accept it. A calendar limit and a small size both point to lot/batch recording
- [ ] **B** Accept it, but add LL with the cycle/usage note text
- [ ] **C** Remove both codes. A part the FAA finds too small to mark is exempt from 45.15(c)
- [ ] **D** Keep SER. Attach the unique serial number to the part or container under 45.15(d)

M4-05 Limits on a lot, and shelf life

An ALS entry normally lists a part number, a limit, and a task. It states the limit against a part.

An entry can state the limit against a manufactured lot/batch of material instead. This is uncommon. Where it happens, the material takes LOT and LL rather than SER.

Do not confuse that case with shelf life. Shelf life and cure date limits also attach to lots. They live in the maintenance manual or the material specification. They do not live in the ALS.

Airworthiness limitation

Published in the ALS. Approved as part of the type design. Drives 45.15(c) marking when it attaches to a discrete part.

Shelf life or cure date

Published in the maintenance manual or the material specification. Attaches to a manufacturing lot/batch. Drives LL and LOT.

Airworthiness limitations and shelf life are separate regimes. Ask where the limit is published before you ask how it is expressed.

CHECK YOURSELF · M4-05

A structural adhesive has a cure date limit stated in its material specification. It has no ALS entry. Which codes apply?

- [ ] **A** SER and LL, because 45.15(c) applies to any life limit
- [ ] **B** LL only, because the material specification controls the limit
- [ ] **C** LL and LOT
- [ ] **D** LOT only, because a cure date is not a life limit

M4-06 Safe-life Principal Structural Elements

BACKGROUND · NOT TESTED · SAFE LIFE AND DAMAGE TOLERANCE

Most transport airplane structure is damage tolerant. It is designed to carry load with a crack present until an inspection finds the crack. Some structure cannot meet that standard. For that structure, the applicant shows it will not crack during its service life, then retires it before that life ends.

DEFINITION

A **Safe-Life Principal Structural Element (PSE)** is a PSE for which the applicant establishes that compliance with the damage-tolerance requirements is impractical. Under 14 CFR 25.571(c), that structure must instead be shown by analysis supported by test evidence to withstand the repeated loads of variable magnitude expected during its service life without detectable cracks.

The design approval holder publishes the safe-life limit in the ALS as a mandatory replacement time. The ALS states it in flight cycles or flight hours. That makes a safe-life PSE a cycle/usage-controlled life-limited item under 14 CFR 43.10.

A safe-life PSE that is a discrete part with an ALS replacement time is serialized under 14 CFR 45.15(c). It takes LL and SER.

Its detail parts follow a different rule. Lot/batch recording applies to the raw material heat and lot of those detail parts. It does not apply to the safe-life PSE part itself.

NOT A PART REPLACEMENT TIME: THE LIMIT OF VALIDITY

The **Limit of Validity (LOV)** bounds the engineering data that supports the airplane's structural maintenance program. It applies to the airplane. It is not a part replacement time.

A safe-life limit retires a part. The LOV bounds the airplane. Do not apply LL to a part because of the LOV.

CHECK YOURSELF · M4-06

A landing gear side brace is a safe-life PSE with an ALS replacement time of 60,000 landings. It is a discrete part. Which codes apply to the brace itself?

- [ ] **A** LOT, because safe-life structure is traced by raw material heat
- [ ] **B** LL and SER
- [ ] **C** LL only, because the ALS already controls the brace
- [ ] **D** SER only, because the Limit of Validity sets its life

M4-07 Damage-tolerant PSEs and constituent materials

A damage-tolerant PSE takes one of two paths. The difference is what the ALS entry attaches to.

Inspection at a location

The ALS entry is a structural inspection at an airframe location. There is no separately manufactured part. No serialization. The program tracks the PSE by airplane and structural location. No code.

Inspection on a discrete part

The PSE is a discrete, separately manufactured part with an ALS inspection interval. 14 CFR 45.15(c) requires serialization. SER.

Constituent materials follow a third path. Composite resins, structural adhesives, and prepregs go into PSEs. They are too small to serialize. Before installation, they are age and shelf-life limited, and they are lot/batch recorded. Once installed, they inherit the cycle and usage life of their higher-level assembly.

PSE case	Codes
Safe-life PSE, discrete part with an ALS replacement time	LL, SER
Safe-life PSE detail parts and constituent materials	LOT
Damage-tolerant PSE, discrete part with an ALS inspection interval	SER
Damage-tolerant PSE, location-based inspection	None

Every row that carries SER or LOT starts a cascade to the end item.

CHECK YOURSELF · M4-07

A composite prepreg with a storage age limit is cured into a safe-life PSE. How is the prepreg's traceability handled?

- [ ] **A** Not recorded, because the PSE's serial number covers everything inside it
- [ ] **B** Given its own cycle/usage LL note and a serial number, because the PSE it goes into is cycle-limited
- [ ] **C** Serialized, because it becomes part of a serialized PSE
- [ ] **D** Lot/batch recorded before installation; once installed it inherits the PSE's cycle and usage life

M4-08 Premium selected materials

BACKGROUND · NOT TESTED · MATERIAL DESIGN VALUES

Material design values set how much stress analysis lets a part carry. 14 CFR 25.613 governs them.

Under 14 CFR 25.613(b), design values must be chosen to minimize the probability of structural failure due to material variability. Some structure carries its load through a single member. If that member fails, the component loses its structural integrity. For that structure, the values must assure material strength with 99 percent probability at 95 percent confidence. Engineers call this the A-basis. Redundant structure can lose one element and shed its load safely to other members. For that structure, the standard is 90 percent probability at 95 percent confidence. Engineers call this the B-basis.

Premium selection lets a program use greater material design values. Under 14 CFR 25.613(e), greater material design values may be used:

14 CFR 25.613(E)

“if a ‘premium selection’ of the material is made in which a specimen of each individual item is tested before use to determine that the actual strength properties of that particular item will equal or exceed those used in design.”

The key words are “each individual item.” Each part is tested. Each result belongs to one part.

Neither 14 CFR 25.613(e) nor AC 25.613-1 names serialization. It is still required in practice. Each test result must tie to one part instance, from specimen through acceptance to installation. Only a distinctive product-unit identifier can hold that tie.

Lot/batch recording does the other half. It captures heat and lot traceability, and a record that correlates each specimen to its part. If the part also carries an ALS entry, 45.15(c) marking applies as well.

A premium selected material part takes SER and LOT.

WHAT CONFIGURATION MANAGEMENT CHECKS

Every ALS entry on a discrete part has SER on its line.

LL appears only where the entry carries a replacement limit.

Each premium selected part has a serial number and a lot record that ties its test specimen to it.

CHECK YOURSELF · M4-08

A stress team raises the allowable on forged fittings. It tests a coupon cut from each forging before the fitting is used. Why must each fitting be serialized?

- [ ] **A** Serialization of each fitting replaces the need for heat and lot records on the forgings
- [ ] **B** Forgings are always cycle-limited, so they are always serialized
- [ ] **C** Each test result must stay tied to one fitting from specimen to installation
- [ ] **D** 14 CFR 25.613(e) states a serialization requirement in its text

M4-DR Parts-list drill

Read each line. Tick every part note code it takes. Then check it.

STANDING ASSUMPTIONS FOR EVERY LINE

Unless the line says otherwise, the item has no children that carry SER or LOT, it is not the end item, and the contract carries no IUID clause.

A line is right only when the set is exact. A missing code is a defect. An extra code is also a defect. Tick **No code** when nothing applies.

LINE M4-L1 · FIND 2 · PART NO. EX-6610-551

BRACE, SIDE, MAIN LANDING GEAR

- Safe-life PSE.
- Discrete part with an ALS replacement time of 60,000 landings.

☐ SER ☐ LOT ☐ TR ☐ IUID ☐ LL ☐ SH ☐ CSI ☐ FC ☐ No code

LINE M4-L2 · FIND 4 · PART NO. EX-8850-012

CYLINDER, OXYGEN, CREW

- Discrete part.
- ALS replacement limit: 15 years from date of manufacture.

☐ SER ☐ LOT ☐ TR ☐ IUID ☐ LL ☐ SH ☐ CSI ☐ FC ☐ No code

LINE M4-L3 · FIND 7 · PART NO. EX-5120-230

FITTING, STABILIZER ATTACH

- Discrete, separately manufactured part.
- ALS entry: inspect every 6,000 flight cycles. No replacement time.

☐ SER ☐ LOT ☐ TR ☐ IUID ☐ LL ☐ SH ☐ CSI ☐ FC ☐ No code

LINE M4-L4 · FIND N/A · PART NO. LOCATION 4-118

WING LOWER SKIN, STRINGER 12 RUNOUT (INSPECTION ZONE)

- ALS entry: inspect the zone every 12,000 flight cycles.
- No separately manufactured part is identified by the entry.

☐ SER ☐ LOT ☐ TR ☐ IUID ☐ LL ☐ SH ☐ CSI ☐ FC ☐ No code

LINE M4-L5 · FIND 3 · PART NO. EX-7730-118

FITTING, WING-TO-BODY, FORGED

- Uses elevated design values under a premium selection.
- A coupon from each forging is tested before use.

☐ SER ☐ LOT ☐ TR ☐ IUID ☐ LL ☐ SH ☐ CSI ☐ FC ☐ No code

LINE M4-L6 · FIND 5 · PART NO. EX-6610-560

LUG, FORGED, DETAIL (INSIDE A SAFE-LIFE PSE ASSEMBLY)

- A detail part of a safe-life PSE assembly.
- It holds no ALS entry of its own.
- Its raw material heat and lot must be traceable.

☐ SER ☐ LOT ☐ TR ☐ IUID ☐ LL ☐ SH ☐ CSI ☐ FC ☐ No code

LINE M4-L7 · FIND 8 · PART NO. EX-8850-044

LUG, INSPECT AND REPLACE

- Discrete part.
- ALS entry: inspect every 5,000 flight cycles and replace at 30,000 flight cycles.

☐ SER ☐ LOT ☐ TR ☐ IUID ☐ LL ☐ SH ☐ CSI ☐ FC ☐ No code

LINE M4-L8 · FIND 9 · PART NO. EX-SKT-ACT-02

SEAL KIT, ACTUATOR

- Shelf-life limit on the manufacturing lot.
- No certification or contract requirement calls for serialization.

☐ SER ☐ LOT ☐ TR ☐ IUID ☐ LL ☐ SH ☐ CSI ☐ FC ☐ No code

M4-EX Exercises

Write your reasoning, not just your conclusion. The reasoning is what is being assessed, and several of these have more than one defensible answer. Where a question turns on a fact you have not been given, **say so, and say what you would ask for and whom you would ask.** Recognizing the missing fact is part of the exercise.

Exercise 1: Reading an Airworthiness Limitations Section

You are given four ALS entries. (1) Main landing gear side brace, a discrete part: replace at 60,000 landings. (2) Wing lower skin at a stringer runout: inspect every 12,000 flight cycles; no separately manufactured part is identified. (3) Crew oxygen cylinder, a discrete part: replace at 15 years. (4) Horizontal stabilizer attach fitting, a discrete part: inspect every 6,000 flight cycles and replace at 45,000 flight cycles.

a. Give the codes for each entry, and the LL note text where LL applies. For each one, say which of the three questions decided it.

b. A reviewer argues that entry 3 should be lot/batch recorded because its limit is calendar time. Answer the reviewer, citing the rule.

c. Entry 4 carries two limits. Explain why only one of them drives LL.

d. Entry 2 takes no code. Explain how that location is still controlled.

Exercise 2: A premium selection program

A stress team wants to raise the design allowable on a set of forged fittings. A test coupon cut from each forging will be tested before the fitting is used. The forgings come from three heats. The fittings go into a wing-to-body joint assembly.

a. Which regulation permits this, and what does it require of each item?

b. The rule does not name serialization. Explain why serialization is still required.

c. What does lot/batch recording add that serialization does not?

d. Which codes go on the fitting? Which codes does the joint assembly take, and why?

e. One coupon record cannot be matched to a fitting. Can that fitting use the higher allowable? Who owns the decision, and what record is missing?

MODULE 5

Lot/batch sources and special handling

Where lot/batch recording requirements come from: castings, process and batch conditions, counterfeit avoidance for electronic parts, and special handling, including explosive and radioactive material

M5-01 Where lot/batch recording comes from

WHAT YOU WILL BE ABLE TO DO

Name the sources that drive lot/batch recording.

Code castings, process items, and electronic parts under a counterfeit avoidance clause.

Write a special handling note a technician can act on.

Place explosive and radioactive material notes on the right drawings.

An O-ring, a casting, and a resistor all take LOT. The O-ring has a shelf life. Quality accepted the casting on the evidence of its batch. The resistor falls under a counterfeit avoidance clause. Three reasons, one code.

LOT drives one action: record lot/batch numbers, or apply and record lot/batch numbers. As with SER, the requirement comes from many places.

Source	Why the lot matters
Age/shelf-life-controlled life-limited items	The limit attaches to the manufacturing lot/batch
ALIs expressed against a manufactured lot/batch (uncommon)	The ALS limit is on the lot
Safe-life PSE detail parts and constituent materials	Raw material heat and lot
Premium selected materials	Heat traceability and specimen correlation
Critical Safety Items	Traceability to an approved source, with lot and heat control
Fracture-Critical Traceable Parts and Fracture-Critical Parts	Fracture control relies on material properties and process condition
Castings and alternative approved material design values	Batch records support reduced inspection or approved values
Process and batch conditions	Identity or performance is set by a batch
Counterfeit avoidance for electronic parts	The source chain back to the original manufacturer
Special handling, including explosive and radioactive material	Handling conditions are recorded as a lot

Recall the two note forms. A raw material callout takes the short LOT text. A detail part or vendor part takes the long one. And recall the location gap: the parent of a lot/batch recorded item is serialized, so every LOT item starts a cascade to the end item.

CHECK YOURSELF · M5-01

Aluminum plate is called out as raw material on a parts list, and its specification requires lot traceability. A vendor O-ring on the same list also requires lot/batch recording. Which note texts apply?

- [ ] **A** No LOT on the plate, because raw material cannot be traced
- [ ] **B** Short LOT text on both, because neither is serialized
- [ ] **C** Long LOT text on both, because both require traceability
- [ ] **D** Short LOT text on the plate, long LOT text on the O-ring

M5-02 Castings and alternative design values

BACKGROUND · NOT TESTED · CASTING FACTORS

A casting's strength varies more than wrought material does. Porosity and inclusions hide inside it. 14 CFR 25.619 requires special factors wherever strength is uncertain, and 14 CFR 25.621 sets the **casting factors** for castings. A casting factor is an extra factor of safety applied to the casting's design loads.

Three conditions in Part 25 depend on production lot and batch records.

- **Non-critical castings with reduced non-visual inspection.** 14 CFR 25.621(d)(5) lets a program inspect fewer castings per production batch by non-visual methods. The program first needs an approved quality control procedure. Quality control must track and record each production batch to support the reduced sampling. See FAA AC 25.621-1.
- **Castings using process monitoring to support a casting factor of 1.0.** 14 CFR 25.621(c)(1) and 14 CFR 25.621(d)(1) allow a casting factor of 1.0 or greater under stated conditions. For critical castings, those conditions include process monitoring. The program tests coupons cut from the prolongations of each casting. Or it tests coupons from each set of castings produced from a single pour into a single mold in a runner system. Records must tie each coupon to its casting or pour. See FAA AC 25.621-1.
- **Alternative approved material design values.** 14 CFR 25.613(f) permits other material design values when the Administrator approves them. The rule does not itself require lot/batch records. The approval carries process and statistical conditions. Production lot and batch records show ongoing conformity to those conditions.

In each case, the batch record earns the relief: the reduced factor, the reduced sampling, or the approved value. Lose the batch record, and the relief goes with it.

Each of these items takes LOT.

CHECK YOURSELF · M5-02

A non-critical casting program samples fewer castings per batch for radiographic inspection under an approved quality control procedure. Why does the casting take LOT?

- [ ] **A** The reduced sampling is only valid if each production batch is tracked and recorded
- [ ] **B** 14 CFR 25.621 requires a serial number on every casting
- [ ] **C** Every casting is serialized, and LOT is its identity code
- [ ] **D** Castings always carry a shelf-life limit that attaches to the pour, so the pour is recorded

M5-03 Process and batch conditions

Some items get their identity or their performance from a batch rather than from a unit. Lot/batch recording applies to:

- Items fabricated using specific batches of bulk materials or processes.
- Items whose testing or certification is conducted by lot/batch.
- Items whose performance may depend on process or batch conditions, including storage conditions, temperature, solution strength, application technique, and cure time.

Common examples:

- Printed wiring boards.
- Heat-treated high-strength material.
- Forged items.
- Specialized metallic fasteners, for heat and lot traceability.
- Parts and materials with specific storage requirements.
- Cured epoxy-filled honeycomb panels.
- Electrical fuses.
- O-rings.
- Filters.
- Simple high-volume electromechanical switches.

The switches are standard toggles, general-purpose push-button switches, circuit breakers, and internal switches used inside non-critical line replaceable units.

This is the widest lot/batch source by item count.

FASTENERS ARE NOT LIFE-LIMITED

Specialized metallic fasteners have no shelf life. They carry heat and lot traceability. They take LOT under this source. They do not take LL.

The pattern to avoid is reasoning from “lot recorded” to “shelf life.” Many lot-recorded items have no life limit at all.

An item can carry this source and a criticality designation at the same time. The lot records are the same records.

CHECK YOURSELF · M5-03

Specialized high-strength fasteners require heat and lot traceability. Which codes apply?

- [ ] **A** SER and LOT
- [ ] **B** LOT
- [ ] **C** LL and LOT
- [ ] **D** LL only

M5-04 Counterfeit avoidance for electronic parts

Counterfeit avoidance drives lot/batch traceability on electronic parts. It is a contract requirement, not a design requirement. It applies when the contract includes either of two clauses:

- **DFARS 252.246-7007**, Contractor Counterfeit Electronic Part Detection and Avoidance System.
- **DFARS 252.246-7008**, Sources of Electronic Parts.

DEFINITION, DFARS 252.246-7007

Electronic part means "an integrated circuit, a discrete electronic component (including, but not limited to, a transistor, capacitor, resistor, or diode), or a circuit assembly."

The definition is broad. It reaches ordinary passive components, not only microcircuits. A resistor is an electronic part.

Both clauses state the traceability requirement the same way. The contractor keeps risk-based processes that let it track electronic parts from the original manufacturer to product acceptance by the Government. That holds whether the parts arrive as discrete parts or inside assemblies.

DFARS 252.246-7008 sets the sourcing order:

1. **The original manufacturer, or its authorized suppliers**
   Or suppliers that obtain the parts only from the original manufacturer or its authorized suppliers.

2. **A contractor-approved supplier**
   Only when the part is not available from the first tier. The contractor then takes responsibility for authenticity, and for inspection, testing, and authentication under the applicable industry standards.

SAE AS5553 and SAE AS6081 are the standards commonly invoked for that inspection, testing, and authentication.

BACKGROUND · NOT TESTED · THE TWO SAE STANDARDS

SAE AS5553 covers counterfeit avoidance, detection, mitigation, and disposition for organizations that buy and integrate electronic parts. SAE AS6081 covers the same ground for distributors.

When the contractor cannot establish traceability, inspection, testing, and authentication replace it. That substitution is expensive. The lot and source record is the cheaper path.

CHECK YOURSELF · M5-04

A contract includes DFARS 252.246-7008. A buyer wants to source a resistor from an independent distributor with no link to the original manufacturer, because it is not a microcircuit. What is wrong?

- [ ] **A** Nothing. The clauses apply only to integrated circuits
- [ ] **B** The clause bans independent distributors under all conditions
- [ ] **C** A resistor is an electronic part, so the sourcing order and traceability apply
- [ ] **D** Nothing, provided the resistor is serialized and inspected at receiving

M5-05 Which source, and which lot

Counterfeit avoidance is commonly mistaken for a serialization requirement. It is not one.

Serialization records

Which unit. Where that unit is installed.

Counterfeit avoidance records

Which source. Which lot. The chain back to the original component manufacturer.

The record structure is lot/batch with a source chain. The obligation is to trace a part back to the company that made it. A serial number applied at receiving says nothing about where the part came from.

An electronic part can carry this source and a shelf-life source at the same time. Electronic components, microchips, capacitors, and diodes appear in both lists. One LOT code satisfies both. The reasons differ. The lot records are the same records.

A counterfeit-controlled electronic part takes LOT. If it also has a shelf-life limit, it takes LL and LOT.

CHECK YOURSELF · M5-05

A capacitor is bought under DFARS 252.246-7007 and has a shelf-life limit on its lot. Which codes does it take?

- [ ] **A** LOT only, because the shelf life is covered by counterfeit control
- [ ] **B** SER and LOT, one identity for each source
- [ ] **C** SER and LL, because counterfeit control needs a serial number
- [ ] **D** LL and LOT

M5-06 Special Handling Required

Special Handling Required items are Lot/Batch Recording Required items. Manufacturing and Production must follow special handling and packaging methods when storing and working with them.

They include:

- Sensitive electronic parts or assemblies.
- Fragile equipment.
- Easily lost parts of high value.
- Perishable items.
- Items that must be stored under specific environmental or security conditions.

A critical limit on any of these factors can call for special handling, packaging, or environmental control:

Environment

Atmospheric pressure. Fungus. Hail. Humidity. Rain. Salt spray. Sand and dust. Solar radiation. Temperature. Wind.

Physical and electrical

Shock. Vibration. Electrostatic discharge sensitive. Sensitive electronic devices.

Material, life, and security

Life limits, including expiration dates. Hazardous material. Security classification.

The build environment records the special handling conditions, including segregation and packaging, as a lot. That is why SH always travels with LOT.

Parts lists containing these items must specify the handling, packaging, or environmental requirements. Parts lists that call out off-the-shelf purchased items must identify the special requirements for those items too. An off-the-shelf part does not arrive with the requirement attached.

CHECK YOURSELF · M5-06

A parts list calls out an off-the-shelf humidity-sensitive sensor. The engineer omits SH, reasoning that the vendor's datasheet covers storage. What is wrong?

- [ ] **A** Nothing. The vendor datasheet satisfies the requirement
- [ ] **B** SH is needed only if the sensor is also a Critical Safety Item
- [ ] **C** Only the LOT code is missing, because SH does not apply to purchased items
- [ ] **D** Parts lists must identify special requirements for off-the-shelf items too

M5-07 Writing the special handling note

For items requiring special handling, apply both notes:

LOT LOT/BATCH RECORDING REQUIRED. THIS ITEM REQUIRES LOT/BATCH TRACEABILITY AND TRACEABILITY MUST BE MAINTAINED BETWEEN THIS ITEM AND ALL HIGHER-LEVEL PHYSICAL ITEMS UP TO THE END ITEM.

SH SPECIAL HANDLING REQUIRED – [ENTER THE SPECIAL HANDLING, PACKAGING, OR ENVIRONMENTAL CONTROL REQUIRED, OR REFERENCE THE SPECIFICATION THAT CONTAINS THAT INFORMATION].

The bracketed field is where the work is. State the control, or reference the specification that states it. Write it so a technician at a bench can act on it without looking anything else up, or can find the specification that tells them.

Three examples:

SH SPECIAL HANDLING REQUIRED – THESE PARTS ARE EXTREMELY FRAGILE. DO NOT DROP OR JAR. PROVIDE 1/8-INCH POLYURETHANE FOAM UNDER THEIR CONTAINER ON THE BENCH OR WHEN STORING OR SHIPPING.

SH SPECIAL HANDLING REQUIRED – TEMPERATURE SHALL BE KEPT BETWEEN 33°F AND 150°F AND RELATIVE HUMIDITY SHALL BE 70 PERCENT MAXIMUM DURING SHIPMENT AND INSTALLATION.

SH SPECIAL HANDLING REQUIRED – ELECTROSTATIC DISCHARGE SENSITIVE. HANDLE, PACKAGE, AND STORE UNDER AN ELECTROSTATIC DISCHARGE CONTROL PROGRAM IN ACCORDANCE WITH ANSI/ESD S20.20.

BACKGROUND · NOT TESTED · CANCELLED STANDARDS IN OLDER NOTES

Older ESD notes cite MIL-STD-1686 and DOD-HDBK-263. MIL-STD-1686C was cancelled on 12 January 2021. The cancellation notice names ANSI/ESD S20.20 as its replacement.

When you revise a parts list that carries an older ESD note, bring the reference current. A note that points at a cancelled standard sends the reader to a document nobody maintains.

CHECK YOURSELF · M5-07

Which special handling note text does its job?

- [ ] **A** SPECIAL HANDLING REQUIRED – TEMPERATURE SHALL BE KEPT BETWEEN 33°F AND 150°F DURING SHIPMENT AND INSTALLATION.
- [ ] **B** SPECIAL HANDLING REQUIRED – SEE VENDOR.
- [ ] **C** SPECIAL HANDLING REQUIRED.
- [ ] **D** SPECIAL HANDLING REQUIRED – HANDLE WITH CARE.

M5-08 Explosive material

Explosive materials include, but are not limited to, ammunition, pyrotechnics, destructors, detonators, initiators, thrusters, JATO units, fuels, catapults, and explosive units. The category includes solids, liquids, and gases.

Before releasing any data, coordinate the use of these materials with the Materials and Processes organization under existing operating procedures. Do the same for any material suspected of being explosive.

For explosive material items, apply:

LOT LOT/BATCH RECORDING REQUIRED. THIS ITEM REQUIRES LOT/BATCH TRACEABILITY AND TRACEABILITY MUST BE MAINTAINED BETWEEN THIS ITEM AND ALL HIGHER-LEVEL PHYSICAL ITEMS UP TO THE END ITEM.

SH EXPLOSIVE MATERIAL – SPECIAL HANDLING REQUIRED – [INCLUDE A DESCRIPTION OF THE METHOD FOR HANDLING EXPLOSIVE MATERIAL AND THE SAFETY PRECAUTIONS TO BE OBSERVED].

The explosive material note text also goes on the drawing picture sheet. If that is not practicable, the picture sheet carries a reference to it. **Every higher-level drawing** carries the note, or a reference to it, too.

Explosive material is the one condition that must be visible at every level above it.

Someone working on a higher-level assembly needs to know what is inside it.

BERYLLIUM IS NOT EXPLOSIVE

Beryllium and beryllium alloys are toxic hazardous materials. They are not explosive materials. Treat them as Special Handling Required items under the hazardous material factor, with SH and LOT.

CHECK YOURSELF · M5-08

A thruster cartridge is installed in a canopy jettison assembly, which goes into the cockpit installation. Where must the explosive material note, or a reference to it, appear?

- [ ] **A** On the cartridge's parts list line only, next to the LOT note
- [ ] **B** On the end item drawing only, where the whole system is shown
- [ ] **C** On the cartridge drawing and the next-higher drawing only
- [ ] **D** On the cartridge's drawing picture sheet and on all higher-level drawings

M5-09 Radioactive material

Radioactive materials are items that Radiation Health Protection has identified as radioactive beyond tolerable limits.

For radioactive material items, apply:

LOT LOT/BATCH RECORDING REQUIRED. THIS ITEM REQUIRES LOT/BATCH TRACEABILITY AND TRACEABILITY MUST BE MAINTAINED BETWEEN THIS ITEM AND ALL HIGHER-LEVEL PHYSICAL ITEMS UP TO THE END ITEM.

SH RADIOACTIVE MATERIAL – SPECIAL HANDLING REQUIRED – WARNING: RADIOACTIVE MATERIAL [IDENTIFY THE ISOTOPE], [NUMBER OF CURIES]. CONTACT RADIATION HEALTH PROTECTION FOR SPECIAL HANDLING.

The note must name the isotope and the amount of radioactivity. A warning without the isotope and the curies is not enough for Radiation Health Protection to act on.

The lowest-level drawing that calls for the radioactive material carries the note. Higher-level drawings carry it when Radiation Health Protection requests it.

	Explosive material	Radioactive material
Codes	SH, LOT	SH, LOT
Who decides it applies	Coordinate with Materials and Processes	Radiation Health Protection identifies it
Note must include	Handling method and safety precautions	Isotope and number of curies
Higher-level drawings	Always, note or reference	When Radiation Health Protection requests it

WHAT CONFIGURATION MANAGEMENT CHECKS

Every SH note states a control or names a specification.

Every SH item also carries LOT.

Explosive material notes appear on every higher-level drawing.

Radioactive material notes name the isotope and the curies.

CHECK YOURSELF · M5-09

A radioactive source is called out on a detail drawing. Radiation Health Protection has not asked for the note on higher assemblies. Where is the note required?

- [ ] **A** On the end item drawing only, as the top-level record
- [ ] **B** On the lowest-level drawing that calls for the material
- [ ] **C** Nowhere, until Radiation Health Protection requests it
- [ ] **D** On every higher-level drawing up to the end item

M5-DR Parts-list drill

Read each line. Tick every part note code it takes. Then check it.

STANDING ASSUMPTIONS FOR EVERY LINE

Unless the line says otherwise, the item has no children that carry SER or LOT, it is not the end item, and the contract carries no IUID clause.

A line is right only when the set is exact. A missing code is a defect. An extra code is also a defect. Tick **No code** when nothing applies.

LINE M5-L1 · FIND 11 · PART NO. EX-3340-900

CASTING, HOUSING, NON-CRITICAL

- Accepted under an approved quality control procedure that reduces non-visual inspection per production batch.

☐ SER ☐ LOT ☐ TR ☐ IUID ☐ LL ☐ SH ☐ CSI ☐ FC ☐ No code

LINE M5-L2 · FIND 15 · PART NO. EX-BLT-HT-12

BOLT, HIGH-STRENGTH, SPECIALIZED

- Requires heat and lot traceability.
- No shelf life.

☐ SER ☐ LOT ☐ TR ☐ IUID ☐ LL ☐ SH ☐ CSI ☐ FC ☐ No code

LINE M5-L3 · FIND 22 · PART NO. EX-RES-0402-10K

RESISTOR, FIXED, FILM

- Contract includes DFARS 252.246-7007 and 252.246-7008.
- No shelf-life limit.

☐ SER ☐ LOT ☐ TR ☐ IUID ☐ LL ☐ SH ☐ CSI ☐ FC ☐ No code

LINE M5-L4 · FIND 23 · PART NO. EX-CAP-TA-47U

CAPACITOR, TANTALUM

- Contract includes DFARS 252.246-7007.
- Shelf-life limit on the manufacturing lot.

☐ SER ☐ LOT ☐ TR ☐ IUID ☐ LL ☐ SH ☐ CSI ☐ FC ☐ No code

LINE M5-L5 · FIND 4 · PART NO. EX-SNS-HUM-2

SENSOR, HUMIDITY-SENSITIVE (OFF THE SHELF)

- Relative humidity must stay at or below 70 percent during shipment and installation.

☐ SER ☐ LOT ☐ TR ☐ IUID ☐ LL ☐ SH ☐ CSI ☐ FC ☐ No code

LINE M5-L6 · FIND 2 · PART NO. EX-CTG-JET-7

CARTRIDGE, THRUSTER, CANOPY JETTISON

- Explosive material.

☐ SER ☐ LOT ☐ TR ☐ IUID ☐ LL ☐ SH ☐ CSI ☐ FC ☐ No code

LINE M5-L7 · FIND 19 · PART NO. EX-BE-CU-172

STRIP, BERYLLIUM COPPER

- Toxic hazardous material.
- Not explosive.

☐ SER ☐ LOT ☐ TR ☐ IUID ☐ LL ☐ SH ☐ CSI ☐ FC ☐ No code

LINE M5-L8 · FIND 3 · PART NO. EX-CTG-FX-55

CARTRIDGE, FIRE EXTINGUISHER, ENGINE

- Explosive material.
- Shelf-life limit on the manufacturing lot.

☐ SER ☐ LOT ☐ TR ☐ IUID ☐ LL ☐ SH ☐ CSI ☐ FC ☐ No code

LINE M5-L9 · FIND 7 · PART NO. EX-RAT-DPL-21

ACTUATOR, RAM AIR TURBINE DEPLOY

- Discrete part.
- ALS entry: replace at 20,000 flight hours.

☐ SER ☐ LOT ☐ TR ☐ IUID ☐ LL ☐ SH ☐ CSI ☐ FC ☐ No code

M5-EX Exercises

Write your reasoning, not just your conclusion. The reasoning is what is being assessed, and several of these have more than one defensible answer. Where a question turns on a fact you have not been given, **say so, and say what you would ask for and whom you would ask.** Recognizing the missing fact is part of the exercise.

Exercise 1: The electronics buy

A program's contract includes DFARS 252.246-7007 and 252.246-7008. A circuit card assembly uses microcircuits, ceramic capacitors, film resistors, and tantalum capacitors with a shelf-life limit. One microcircuit is obsolete and unavailable from the original manufacturer or its authorized distributors. The buyer finds stock at an independent distributor.

a. Which items on the card fall under the counterfeit avoidance clauses? Explain using the definition.

b. Which codes do the ceramic capacitors take, and which do the tantalum capacitors take? Explain the difference.

c. What must happen before the obsolete microcircuit can be bought from the independent distributor, and who carries the responsibility?

d. A colleague suggests serializing each microcircuit at receiving to satisfy the clause. Answer them.

e. Which codes does the circuit card assembly take, and why? Could it take any code beyond the ones its children drive? Say what fact would decide it.

Exercise 2: Handling notes on a parts list

An avionics installation parts list includes five items. A fragile optical sensor, bought off the shelf. An electrostatic discharge sensitive circuit card. A thruster cartridge for an emergency release. A beryllium copper spring. A sealed source in a smoke detector. Radiation Health Protection has identified the sealed source as radioactive beyond tolerable limits. The existing note on the card reads: SH – SPECIAL HANDLING REQUIRED – ESD SENSITIVE.

a. Write the SH note for the fragile optical sensor. Explain what makes it actionable.

b. What is wrong with the existing note on the circuit card, and what would you write instead?

c. Which of these items need a note or reference on higher-level drawings? Explain the rule for each.

d. A reviewer treats the beryllium copper spring as explosive material. Correct them, and say what changes in the notes.

MODULE 6

Item Unique Identification

When the contract requires IUID, what a UII holds, how to mark and report it, and how to code each item

M6-01 IUID is a contract requirement

WHAT YOU WILL BE ABLE TO DO

Decide whether the contract requires IUID for an item.

Name the data elements in a UII.

Code an IUID item, a parent item, and an end item.

Say when and how the contractor reports UIIs.

Two identical hydraulic pumps leave the same factory. One goes to an airline. One goes to the Air Force. Only the second may need a data matrix mark and a registry record. The difference is the contract.

The Item Unique Identification Required code, IUID, drives two actions. Mark the item with a unique item identifier (UII). Report that UII to the Department of Defense IUID Registry.

IUID comes from the contract, not from the type design. It applies when the contract includes **DFARS 252.211-7003**, Item Unique Identification and Valuation.

- It does not apply to aircraft or subsystems delivered only under FAA certification, unless the contract requires it.
- When the prime contract requires IUID for an item that a supplier provides, the contractor flows DFARS 252.211-7003 down to that subcontract. That includes subcontracts for commercial products and commercial services.

Every UII carries a serial number, so an IUID item is a serialized item. The UII adds two things a serial number alone does not: a machine-readable mark, and a registry record.

That is why the IUID code travels with SER. The end item is the one exception. It never takes SER, because the contract assigns its identity. Everywhere below the end item, SER brings the cascade with it.

CHECK YOURSELF · M6-01

A commercial transport airplane is delivered under FAA certification only. The contract has no DFARS clauses. A cycle-limited actuator on it costs $12,000. Does it take IUID?

- [ ] **A** No. IUID comes from DFARS 252.211-7003 in the contract, which is absent
- [ ] **B** Yes. A high-value item takes IUID whatever the contract says
- [ ] **C** Only if the actuator is a Line Replaceable Unit removed on the flight line
- [ ] **D** Yes. Every cycle-limited part takes IUID

M6-02 IUID definitions

Look at a data matrix mark on a military part. It holds a short string of data. That string names the enterprise that assigned the identity and gives the item's serial number. Sometimes it also gives the original part, lot, or batch number.

DFARS 252.211-7003 defines the key terms. They are used as written.

DOD ITEM UNIQUE IDENTIFICATION

"a system of marking items delivered to DoD with unique item identifiers that have machine-readable data elements to distinguish an item from all other like and unlike items. For items that are serialized within the enterprise identifier, the unique item identifier shall include the data elements of the enterprise identifier and a unique serial number. For items that are serialized within the part, lot, or batch number within the enterprise identifier, the unique item identifier shall include the data elements of the enterprise identifier; the original part, lot, or batch number; and the serial number."

UNIQUE ITEM IDENTIFIER

"a set of data elements marked on items that is globally unique and unambiguous. The term includes a concatenated unique item identifier or a DoD recognized unique identification equivalent."

PARENT ITEM

"the item assembly, intermediate component, or subassembly that has an embedded item with a unique item identifier or DoD recognized unique identification equivalent."

Read the first definition for its data elements. There is always an **enterprise identifier**, which says who made or assigned the identity. There is always a **serial number**. Sometimes there is an **original part, lot, or batch number** in between. Every combination ends in a serial number.

"Like and unlike items" matters. The UII must be unique against every other item, not only against other items of the same part number. That is what "globally unique" means.

CHECK YOURSELF · M6-02

A UII built within a lot number contains which data elements?

- [ ] **A** The part number, the contract number, and the delivery order number
- [ ] **B** The enterprise identifier, the original lot or batch number, and a serial number
- [ ] **C** The lot or batch number and the part number only
- [ ] **D** The enterprise identifier and the lot or batch number only

M6-03 When IUID applies

A $900 valve on a defense contract takes IUID. The item manager serially manages it. A $12,000 actuator on an FAA-only program does not. The contract decides.

DFARS 211.274-2(a) sets the policy. The contract identifies the items.

Category	When IUID is required	Authority
At or above the value threshold	The Government's unit acquisition cost is $5,000 or more. The contract line item sets the unit acquisition cost.	DFARS 211.274-2(a)(1)
Below the value threshold	Unit acquisition cost under $5,000, when the requiring activity determines IUID is required for mission essential or controlled inventory items. The contract identifies them.	DFARS 211.274-2(a)(2)
DoD serially managed items	Regardless of value. Includes embedded subassemblies, components, and parts. Embedded items are listed in the attachment called out by DFARS 252.211-7003(c)(1)(iii).	DFARS 211.274-2(a)(3)(i)
Parent items	Regardless of value. At delivery the contractor reports the parent's UII with the UII of each embedded item, per DFARS 252.211-7003(e).	DFARS 211.274-2(a)(3)(ii)
Warranted serialized items	Regardless of value, for a serialized item delivered under a warranty.	DFARS 211.274-2(a)(3)(iii)
Special tooling and special test equipment	Regardless of value, for a major defense acquisition program item designated for preservation and storage under section 815 of the FY2009 NDAA (Public Law 110-417).	DFARS 211.274-2(a)(3)(iv)
High risk items	Regardless of value, when the requiring activity identifies the item as vulnerable to supply chain threat, a target of cyber threats, or susceptible to counterfeiting.	DFARS 211.274-2(a)(3)(v)

A **DoD serially managed item** is one the primary inventory control activity identifies as serially managed under DoDI 8320.04. These include:

- Capital asset equipment.
- Critical Safety Items.
- Relocatable facilities.
- Special test equipment and special tooling that is part of a major defense acquisition program.
- Controlled inventory items, as appropriate.
- Depot-level repairable items, as determined by the item manager or the program manager.

The requirement covers reparable and nonreparable items.

Value is only the first gate. Most of the list applies regardless of value.

CHECK YOURSELF · M6-03

A $900 depot-level repairable valve is identified by the item manager as serially managed. The contract includes DFARS 252.211-7003. Does it take IUID?

- [ ] **A** Yes. DoD serially managed items require IUID regardless of value
- [ ] **B** Only if it is also a parent item
- [ ] **C** No. It is below the $5,000 threshold
- [ ] **D** Only if the design control activity also designates it a Critical Safety Item

M6-04 Marking, construct, and reporting

Marking. The build environment marks the UII data elements on the item. The mark is a two-dimensional data matrix symbol that complies with ISO/IEC 16022, ECC200. Marking follows MIL-STD-130, Identification Marking of U.S. Military Property, latest version.

- The build environment verifies each mark with an automatic identification technology device programmed to the requirements of Appendix A of MIL-STD-130.
- The issuing agency code is not placed on the item.
- When an item is too small or otherwise impractical to mark, follow the alternative marking provisions of MIL-STD-130.
- Label shipments, storage containers, and packages that contain uniquely identified items in accordance with MIL-STD-129, Military Marking for Shipment and Storage, latest version.

Serialization construct. The enterprise chooses one of three methods.

Within the enterprise identifier

Serial numbers are unique across everything the enterprise makes. MIL-STD-130 calls this Construct 1.

Within the original part, lot, or batch number

Serial numbers are unique within each part, lot, or batch number. MIL-STD-130 calls this Construct 2.

A DoD recognized equivalent

An existing unique identification system the DoD accepts in place of a constructed UII.

Each method gives every item its own identity. **Lot/batch recording alone does not satisfy IUID.**

Reporting. The contractor reports each UII at delivery, as part of the Material Inspection and Receiving Report.

- The contractor reports end items through the receiving report capability in Wide Area WorkFlow (WAWF).

- It reports embedded items through the WAWF embedded items capability, by direct data submission to the IUID Registry, or as a WAWF deliverable attachment.
- For each embedded item, the report includes the UII of the parent item that contains it.

Configuration Management reconciles the IUID code on the design against the UIIs reported.

CHECK YOURSELF · M6-04

A supplier proposes to satisfy IUID on a batch of identical brackets by marking each with the same lot number in a data matrix. What is wrong?

- [ ] **A** Brackets are too small for data matrix marking, so a tag must be used instead
- [ ] **B** Every UII must include a serial number that gives each item its own identity
- [ ] **C** Nothing, provided the data matrix complies with ISO/IEC 16022
- [ ] **D** The mark must also include the issuing agency code in the data matrix

M6-05 Applying the IUID code

For each item below the end item that holds an IUID requirement, apply:

IUID ITEM UNIQUE IDENTIFICATION REQUIRED. MARK THIS ITEM WITH A UNIQUE ITEM IDENTIFIER (UII) IN ACCORDANCE WITH MIL-STD-130 AND REPORT THE UII IN ACCORDANCE WITH DFARS 252.211-7003.

SER SERIALIZATION RECORDING REQUIRED. THIS ITEM IS SERIALIZED AND TRACEABILITY MUST BE MAINTAINED BETWEEN THIS ITEM AND ALL HIGHER-LEVEL PHYSICAL ITEMS UP TO THE END ITEM.

For each parent item that contains an embedded item carrying a UII, apply:

IUID ITEM UNIQUE IDENTIFICATION REQUIRED. THIS ITEM IS A PARENT ITEM. MARK THIS ITEM WITH A UNIQUE ITEM IDENTIFIER (UII) IN ACCORDANCE WITH MIL-STD-130 AND REPORT THE UIIS OF ITS EMBEDDED ITEMS IN ACCORDANCE WITH DFARS 252.211-7003.

Coordinate the parent item level with the contracting activity. The contract governs which assembly is the parent item.

A parent item also takes SER, and it takes the **long** SER text. Parent items require IUID regardless of value, and every UII carries a serial number. So a contract-designated parent item holds a serialization requirement of its own. It is not serialized only because of the cascade:

SER SERIALIZATION RECORDING REQUIRED. THIS ITEM IS SERIALIZED AND TRACEABILITY MUST BE MAINTAINED BETWEEN THIS ITEM AND ALL HIGHER-LEVEL PHYSICAL ITEMS UP TO THE END ITEM.

The end item is the one exception. It never takes SER. The next page covers it.

SER and TR flow up the product tree. IUID does not.

Apply IUID only to items that hold an IUID requirement and to the parent items the contract identifies.

Put the rules side by side on one assembly. A parent item contains an embedded item with a UII. The contract names it as the parent item, so it takes IUID with the parent item text and SER with the long text. Its embedded item carries SER, so it also has a traceable child and takes TR. It takes IUID, SER, and TR.

Its own physical parent, one level up, is different. That assembly is a physical parent of a serialized item, so the cascade gives it SER and TR, in the short texts. Unless the contract names it too, it takes no IUID.

CHECK YOURSELF · M6-05

An avionics unit is identified by the contract as the parent item for an embedded DoD serially managed circuit card. The unit goes into an equipment rack, which goes into the aircraft. Which codes does the rack take, assuming the contract does not name it?

- [ ] **A** No code
- [ ] **B** SER and TR
- [ ] **C** IUID, SER, and TR
- [ ] **D** IUID only

M6-06 The end item and IUID

A defense contract buys an airplane. The contract requires IUID for the airplane. Its parts list line already carries TR with the terminator text. What else does it take?

It takes the IUID code. It does not take SER.

The end item never takes SER. The contract assigns its identity, such as a tail number or a system serial number. An IUID requirement does not change that.

Pick the end item's IUID note text from the two on the previous page. Use the item note text. Use the parent item note text when the contract names the end item as a parent item.

TR TRACEABILITY BEYOND THIS LEVEL IS NOT REQUIRED.

IUID ITEM UNIQUE IDENTIFICATION REQUIRED. MARK THIS ITEM WITH A UNIQUE ITEM IDENTIFIER (UII) IN ACCORDANCE WITH MIL-STD-130 AND REPORT THE UII IN ACCORDANCE WITH DFARS 252.211-7003.

The end item and a parent item, side by side

A contract-designated parent item and an IUID end item both hold an IUID requirement. Only one takes SER.

	Contract-designated parent item	End item
Where it sits	Inside the tree, below the end item	At the top of the tree
IUID note text	Parent item text	Item text, or parent item text when the contract names it a parent item
SER	Long text. It holds its own IUID requirement.	None. The contract assigns its identity.
TR	Short text, when it has a traceable child	Terminator text

WHAT CONFIGURATION MANAGEMENT CHECKS

Every IUID item below the end item also carries SER.

Each UII the contractor reports matches an IUID code on the design.

The contractor reports each embedded item with the UII of its parent item.

The end item carries no SER.

CHECK YOURSELF · M6-06

A missile system is the end item. The contract requires IUID for it and does not name it a parent item. Which codes does the missile take?

- [ ] **A** SER and TR, in the short texts
- [ ] **B** TR (terminator text), IUID (item text), and SER (long text)
- [ ] **C** TR (terminator text) and IUID (item text)
- [ ] **D** IUID (parent item text) and SER (long text)

M6-DR Parts-list drill

Read each line. Tick every part note code it takes. Then check it.

STANDING ASSUMPTIONS FOR EVERY LINE

Unless the line says otherwise, the item has no children that carry SER or LOT, it is not the end item, and the contract carries no IUID clause.

A line is right only when the set is exact. A missing code is a defect. An extra code is also a defect. Tick **No code** when nothing applies.

LINE M6-L1 · FIND 6 · PART NO. EX-6200-310

VALVE, DEPOT-LEVEL REPARABLE

- The contract includes DFARS 252.211-7003.
- The item manager identifies it as serially managed.
- Unit acquisition cost: $900.

☐ SER ☐ LOT ☐ TR ☐ IUID ☐ LL ☐ SH ☐ CSI ☐ FC ☐ No code

LINE M6-L2 · FIND 3 · PART NO. EX-AV-2210

UNIT, AVIONICS (CONTRACT-DESIGNATED PARENT ITEM)

- The contract includes DFARS 252.211-7003 and names this unit as the parent item.
- It contains an embedded circuit card that carries a UII.

☐ SER ☐ LOT ☐ TR ☐ IUID ☐ LL ☐ SH ☐ CSI ☐ FC ☐ No code

LINE M6-L3 · FIND 4 · PART NO. EX-RK-0900

RACK, EQUIPMENT

- Contains the contract-designated avionics parent item.
- The contract does not name the rack.

☐ SER ☐ LOT ☐ TR ☐ IUID ☐ LL ☐ SH ☐ CSI ☐ FC ☐ No code

LINE M6-L4 · FIND 5 · PART NO. EX-ACT-7710

ACTUATOR, CYCLE-LIMITED

- Life-limited by flight cycles.
- FAA certification only. No DFARS clauses in the contract.
- Unit cost: $12,000.

☐ SER ☐ LOT ☐ TR ☐ IUID ☐ LL ☐ SH ☐ CSI ☐ FC ☐ No code

LINE M6-L5 · FIND N/A · PART NO. EX-TAIL-ASSIGNED

AIRPLANE, COMPLETE (END ITEM), DEFENSE CONTRACT

- The top-level asset under contract.
- The contract requires IUID for the airplane and does not name it a parent item.
- Traceable items are installed beneath it.

☐ SER ☐ LOT ☐ TR ☐ IUID ☐ LL ☐ SH ☐ CSI ☐ FC ☐ No code

LINE M6-L6 · FIND 12 · PART NO. EX-CST-BKT-03

CASTING, BRACKET, PROCESS-MONITORED

- A critical casting.
- The program tests coupons cut from the prolongation of each casting to support a casting factor of 1.0.

☐ SER ☐ LOT ☐ TR ☐ IUID ☐ LL ☐ SH ☐ CSI ☐ FC ☐ No code

M6-EX Exercises

Write your reasoning, not just your conclusion. The reasoning is what is being assessed, and several of these have more than one defensible answer. Where a question turns on a fact you have not been given, **say so, and say what you would ask for and whom you would ask.** Recognizing the missing fact is part of the exercise.

Exercise 1: Follow the cascade

A missile system is the end item. Beneath it are five items. A guidance section installation drawing. A guidance section assembly. An inertial measurement unit (IMU). A circuit card inside the IMU, which uses tantalum capacitors with a shelf-life limit. A thermal battery that is explosive material. The contract includes DFARS 252.211-7003. The IMU is a DoD serially managed item. The contract names the guidance section assembly as the parent item for the IMU.

a. List every item in the tree from the bottom up, with its codes and note form.

b. Which items take IUID? Explain why the others do not.

c. The guidance section installation drawing: which codes, and why no SER?

d. What does the end item take? What gives it its identity?

e. Which condition codes appear in the tree, and why don't they cascade? What is different about the explosive material note?

Exercise 2: The IUID challenge

A supplier builds a $3,200 hydraulic pump for a military transport. The prime contract includes DFARS 252.211-7003. The pump is not on the list of serially managed items, and the contract does not identify it as mission essential or controlled inventory. The supplier's subcontract is a commercial products subcontract. The supplier plans to mark each pump with its part number and a lot number in a data matrix, and to report nothing.

a. Does the pump require IUID? Walk through the categories and say what fact you would need to confirm your answer.

b. The supplier argues that a commercial products subcontract is exempt from the IUID clause. Answer them.

c. If IUID did apply, what is wrong with the supplier's marking plan?

d. If IUID did apply, what UII reporting does the contractor owe the Government for the pump, when, and through what channel? Whom would you ask what the subcontract requires of the supplier?

MODULE 7

Critical items

Critical Safety Items, Fracture-Critical Traceable Parts, Fracture-Critical Parts, and Safety-Critical Items: who designates each one, and which codes each one drives

M7-01 Critical items and Critical Safety Items

WHAT YOU WILL BE ABLE TO DO

Tell the four critical designations apart.

Code each one, and name who designates it.

Say which codes cascade and which stay on the item.

Picture three parts. A flight control bellcrank. A wing attach lug with a single load path. A hydraulic shutoff valve. Each one can carry a critical label. The labels sound alike. They come from different authorities, and they drive different codes.

“Critical items” is a broad term. It covers many part classifications related to safety. Some come from customer requirements or regulatory agencies. Others rest on engineering governance and analysis. Three matter for part note codes: Critical Safety Items, Fracture-Critical Traceable Parts, and Safety-Critical Items. The names are close. The rules are not.

DFARS 209.270-2 defines an aviation critical safety item as a part, an assembly, installation equipment, launch equipment, recovery equipment, or support equipment for an aircraft or aviation weapon system if the part, assembly, or equipment contains a characteristic any failure, malfunction, or absence of which could cause any of the following:

DFARS 209.270-2

“(1) A catastrophic or critical failure resulting in the loss of or serious damage to the aircraft or weapon system;”

“(2) An unacceptable risk of personal injury or loss of life; or”

“(3) An uncommanded engine shutdown that jeopardizes safety.”

Section 802 of Public Law 108-136 is the statutory basis. DFARS 209.270 and DFARS PGI 209.270 set the qualification requirements, source approval, and procedures. The contract clause at DFARS 252.209-7010 carries the definitions and the heightened quality assurance surveillance requirement.

Who designates. The design control activity designates Critical Safety Items. DFARS 252.209-7010 defines it, for an aviation critical safety item, as “the systems command of a military department that is specifically

responsible for ensuring the airworthiness of an aviation system or equipment, in which an aviation critical safety item is to be used." System Safety and engineering support provide inputs. They do not make the designation.

Where it does not apply. Critical Safety Item identification does not apply to aircraft or subsystems bought, operated, and maintained under FAA certification. The United States Air Force Engineering Support Activity can still require it. Critical Safety Item policies do not override or duplicate FAA approaches when those approaches apply and meet military Critical Safety Item objectives.

CHECK YOURSELF · M7-01

An engineer's hazard analysis shows a bracket failure could cause loss of the aircraft. The engineer applies the CSI code on their own authority. What is wrong?

- [ ] **A** The design control activity designates Critical Safety Items. Engineering provides inputs
- [ ] **B** System Safety designates Critical Safety Items from its hazard analysis
- [ ] **C** Nothing. Any engineer may designate a CSI from a hazard analysis
- [ ] **D** CSI applies only to engines and rotating hardware, never to airframe structure

M7-02 Coding a Critical Safety Item

Serialization reaches a Critical Safety Item through IUID, in three steps:

1. **DoDI 8320.04**
   Directs the primary inventory control activity to identify Critical Safety Items as DoD serially managed items.

2. **DFARS 211.274-2(a)(3)(i)**
   Requires IUID for any DoD serially managed item, regardless of value.

3. **The UII**
   Every UII carries a serial number. The CSI is serialized.

Lot/batch recording is also required, with traceability to an approved source. Lot and heat control is typical for structural and rotating hardware.

For Critical Safety Items, apply all four:

CSI CRITICAL SAFETY ITEM.

SER SERIALIZATION RECORDING REQUIRED. THIS ITEM IS SERIALIZED AND TRACEABILITY MUST BE MAINTAINED BETWEEN THIS ITEM AND ALL HIGHER-LEVEL PHYSICAL ITEMS UP TO THE END ITEM.

LOT LOT/BATCH RECORDING REQUIRED. THIS ITEM REQUIRES LOT/BATCH TRACEABILITY AND TRACEABILITY MUST BE MAINTAINED BETWEEN THIS ITEM AND ALL HIGHER-LEVEL PHYSICAL ITEMS UP TO THE END ITEM.

IUID ITEM UNIQUE IDENTIFICATION REQUIRED. MARK THIS ITEM WITH A UNIQUE ITEM IDENTIFIER (UII) IN ACCORDANCE WITH MIL-STD-130 AND REPORT THE UII IN ACCORDANCE WITH DFARS 252.211-7003.

CSI does not flow up the product tree. SER and TR cascade to the end item. IUID goes only to the parent items the contract identifies. CSI stays on the item that holds it.

CHECK YOURSELF · M7-02

A flight control bellcrank is designated a Critical Safety Item. It goes into a control linkage assembly. What does the linkage assembly take, assuming the contract does not name it as a parent item?

- [ ] **A** CSI, SER, LOT, IUID, and TR
- [ ] **B** CSI, SER, and TR
- [ ] **C** No code, because only the bellcrank is critical
- [ ] **D** SER and TR

M7-03 Fracture-Critical Traceable Parts and Fracture-Critical Parts

Two fittings hold a wing on. The first carries its load through a single load path. If it cracks, nothing else picks up the load. The second shares its load with other members, but the program gives it special emphasis. Both are safety-of-flight parts. The standard puts them in different categories.

MIL-STD-1530D with Change 1 governs the Aircraft Structural Integrity Program (ASIP). It defines two separate categories. MIL-STD-1530C carries the same definitions.

Fracture-Critical Traceable Part

A safety-of-flight structural component that is either single load path or sized by durability or damage tolerance requirements.

Fracture-Critical Part

A safety-of-flight structural component that is not single load path and is not sized by durability or damage tolerance requirements, but that requires special emphasis due to the criticality.

The two definitions exclude each other. The standard's Figure 1 shows them as separate branches. A Fracture-Critical Traceable Part is not a subclass of a Fracture-Critical Part.

Fracture-Critical Traceable Parts require serialization and lot/batch recording as one of their controls. Lot/batch recording supports material and process traceability. They take FC, SER, and LOT.

FC	FRACTURE-CRITICAL ITEM.
SER	SERIALIZATION RECORDING REQUIRED. THIS ITEM IS SERIALIZED AND TRACEABILITY MUST BE MAINTAINED BETWEEN THIS ITEM AND ALL HIGHER-LEVEL PHYSICAL ITEMS UP TO THE END ITEM.
LOT	LOT/BATCH RECORDING REQUIRED. THIS ITEM REQUIRES LOT/BATCH TRACEABILITY AND TRACEABILITY MUST BE MAINTAINED BETWEEN THIS ITEM AND ALL HIGHER-LEVEL PHYSICAL ITEMS UP TO THE END ITEM.

Fracture-Critical Parts still take lot/batch recording. Fracture control relies on material properties and process condition. Raw material heat and lot traceability applies, with process control on forging, machining, and surface treatment. They take FC and LOT. They do not take SER on this basis.

FC	FRACTURE-CRITICAL ITEM.
LOT	LOT/BATCH RECORDING REQUIRED. THIS ITEM REQUIRES LOT/BATCH TRACEABILITY AND TRACEABILITY MUST BE MAINTAINED BETWEEN THIS ITEM AND ALL HIGHER-LEVEL PHYSICAL ITEMS UP TO THE END ITEM.

NOT THE SAME AS FRACTURE CRITICAL CATEGORY I AND II

Fracture Critical Category I and Category II Parts are a different scheme. They go through the Critical Safety Item or Safety-Critical Item screening process, and that process decides whether part traceability applies.

Fracture-Critical Traceable Parts always require serialization and lot/batch recording. See the critical part and process selection and controls section of MIL-STD-1530D with Change 1, and the Durability and Damage Tolerance Control Plan.

FC does not flow up the product tree. SER and TR do.

CHECK YOURSELF · M7-03

A redundant, non-damage-tolerance-sized structural fitting is a safety-of-flight component that the program says needs special emphasis. Which codes apply on this basis?

- [ ] **A** FC and LOT
- [ ] **B** FC only
- [ ] **C** FC, SER, and LOT
- [ ] **D** SER and LOT

M7-04 Safety-Critical Items

DEFINITION, MIL-STD-882E

A **Safety-Critical Item (SCI)** is “a hardware or software item that has been determined through analysis to potentially contribute to a hazard with Catastrophic or Critical mishap potential, or that may be implemented to mitigate a hazard with Catastrophic or Critical mishap potential.”

BACKGROUND · NOT TESTED · MIL-STD-882E SEVERITY CATEGORIES

MIL-STD-882E defines the severity categories behind those words. **Catastrophic** could result in death, permanent total disability, irreversible significant environmental impact, or monetary loss of $10 million or more. **Critical** could result in permanent partial disability, injuries or occupational illness that may result in hospitalization of at least three personnel, reversible significant environmental impact, or monetary loss of $1 million or more but less than $10 million.

Safety-Critical Items are often confused with Critical Safety Items. MIL-STD-882E settles the point. It states that its definition of “safety-critical item” is independent of the definition of “critical safety item” in Public Laws 108-136 and 109-364.

Critical Safety Item

DFARS and statute. Designated by the design control activity. Takes CSI, SER, LOT, and IUID.

Safety-Critical Item

MIL-STD-882E system safety analysis. Drives design and process controls. No part note code.

No part note code identifies a Safety-Critical Item.

System Safety analysis outputs drive design and process controls for SCIs. They do not drive item marking.

SCI designation does not drive serialization or lot/batch recording on its own. An SCI is serialized or lot/batch recorded when it is life-limited or otherwise holds a serialization or lot/batch recording requirement. Code it from those sources.

WHAT CONFIGURATION MANAGEMENT CHECKS

CSI and FC stay on the item that holds them.

A Safety-Critical Item carries no code on that basis.

Each Critical Safety Item also carries SER, LOT, and IUID.

CHECK YOURSELF · M7-04

A hazard analysis identifies a hydraulic valve as a Safety-Critical Item. No other source applies to it. Which codes does it take?

- [ ] **A** No code on this basis
- [ ] **B** SER and LOT
- [ ] **C** CSI, SER, LOT, and IUID
- [ ] **D** LL and SER

M7-DR Parts-list drill

Read each line. Tick every part note code it takes. Then check it.

STANDING ASSUMPTIONS FOR EVERY LINE

Unless the line says otherwise, the item has no children that carry SER or LOT, it is not the end item, and the contract carries no IUID clause.

A line is right only when the set is exact. A missing code is a defect. An extra code is also a defect. Tick **No code** when nothing applies.

LINE M7-L1 · FIND 5 · PART NO. EX-BCK-FLT-22

BELLCRANK, FLIGHT CONTROL

- Designated a Critical Safety Item by the design control activity.
- The contract includes DFARS 252.211-7003.

☐ SER ☐ LOT ☐ TR ☐ IUID ☐ LL ☐ SH ☐ CSI ☐ FC ☐ No code

LINE M7-L2 · FIND 6 · PART NO. EX-FTG-SLP-40

FITTING, WING ATTACH, SINGLE LOAD PATH

- A safety-of-flight structural component.
- Single load path.

☐ SER ☐ LOT ☐ TR ☐ IUID ☐ LL ☐ SH ☐ CSI ☐ FC ☐ No code

LINE M7-L3 · FIND 7 · PART NO. EX-FTG-RDN-41

FITTING, REDUNDANT, SPECIAL EMPHASIS

- A safety-of-flight structural component.
- Not single load path. Not sized by durability or damage tolerance.
- Requires special emphasis due to criticality.

☐ SER ☐ LOT ☐ TR ☐ IUID ☐ LL ☐ SH ☐ CSI ☐ FC ☐ No code

LINE M7-L4 · FIND 9 · PART NO. EX-VLV-HYD-88

VALVE, HYDRAULIC SHUTOFF

- Identified as a Safety-Critical Item by MIL-STD-882E hazard analysis.
- No other source applies.

☐ SER ☐ LOT ☐ TR ☐ IUID ☐ LL ☐ SH ☐ CSI ☐ FC ☐ No code

LINE M7-L5 · FIND 10 · PART NO. EX-VLV-HYD-89

VALVE, HYDRAULIC SHUTOFF, LIFE-LIMITED

- Identified as a Safety-Critical Item by hazard analysis.
- Also life-limited to 40,000 operating cycles.

☐ SER ☐ LOT ☐ TR ☐ IUID ☐ LL ☐ SH ☐ CSI ☐ FC ☐ No code

LINE M7-L6 · FIND 2 · PART NO. EX-LNK-FLT-20

LINKAGE ASSY, FLIGHT CONTROL

- Contains the Critical Safety Item bellcrank.
- The contract does not name it as a parent item.
- No requirement of its own.

☐ SER ☐ LOT ☐ TR ☐ IUID ☐ LL ☐ SH ☐ CSI ☐ FC ☐ No code

LINE M7-L7 · FIND 4 · PART NO. EX-PMP-HYD-32

PUMP, HYDRAULIC, WARRANTED

- The contract includes DFARS 252.211-7003.
- The pump is a serialized item delivered under a warranty.
- Unit acquisition cost: $3,200.

☐ SER ☐ LOT ☐ TR ☐ IUID ☐ LL ☐ SH ☐ CSI ☐ FC ☐ No code

M7-EX Exercises

Write your reasoning, not just your conclusion. The reasoning is what is being assessed, and several of these have more than one defensible answer. Where a question turns on a fact you have not been given, **say so, and say what you would ask for and whom you would ask.** Recognizing the missing fact is part of the exercise.

Exercise 1: Three critical words

A new program has three items under review. A landing gear drag brace that the design control activity has designated a Critical Safety Item. A wing attach fitting that is single load path and a safety-of-flight structural component. An avionics power supply that the hazard analysis identifies as a Safety-Critical Item. The contract includes DFARS 252.211-7003. A colleague proposes to treat all three the same way, because “they are all critical.”

a. Give the codes for each item and name the authority that drives each one.

b. Explain to the colleague why the three items are coded differently.

c. The drag brace is also cycle-limited. What changes in its codes?

d. The power supply later gets a shelf-life limit on a battery pack inside it. Which item takes new codes, and which ones?

e. The drag brace's physical parent takes SER and TR. It takes none of CSI, LOT, IUID, or LL. Explain why SER and TR cascade and the others stay on the brace.

MODULE 8

Putting it together

The application matrix, the common misreads, and how Configuration Management reconciles the codes against the build records

M8-01 The application matrix

WHAT YOU WILL BE ABLE TO DO

Code a full parts list from every source.

Compare design codes with build records.

Name each gap and route it to its owner.

Run the release check.

Start here when you know the item and need the codes. The matrix is a summary of every rule in this course. Every item carrying SER or LOT also cascades to the end item.

Item or condition	Codes to apply	Taught in
Cycle/usage-controlled life-limited item	LL, SER	Module 3
Age/shelf-life-controlled life-limited item, limit on the lot/batch	LL, LOT	Module 3
ALI on a discrete part, replacement limit in cycles, landings, or flight hours	SER, LL (cycle/usage text)	Module 4
ALI on a discrete part, replacement limit in calendar time	SER, LL (age and shelf-life text)	Module 4
ALI on a discrete part, inspection interval or related procedure only	SER	Module 4
ALI at an airframe location or zone, no separately manufactured part	None	Module 4
ALI expressed against a manufactured lot/batch (uncommon)	LOT, LL	Module 4
Safe-life PSE, discrete part with an ALS replacement time	LL, SER	Module 4
Safe-life PSE detail parts and constituent materials	LOT	Module 4
Damage-tolerant PSE, discrete part with an ALS inspection interval	SER	Module 4
Damage-tolerant PSE, location-based inspection	None	Module 4
Premium selected material part	SER, LOT	Module 4
Critical Safety Item	CSI, SER, LOT, IUID	Module 7

Item or condition	Codes to apply	Taught in
Fracture-Critical Traceable Part	FC, SER, LOT	Module 7
Fracture-Critical Part (not traceable)	FC, LOT	Module 7
Safety-Critical Item	None on this basis	Module 7
Casting with reduced non-visual inspection	LOT	Module 5
Casting using process monitoring for a factor of 1.0	LOT	Module 5
Part using alternative approved material design values	LOT	Module 5
Item established by a process or batch condition	LOT	Module 5
Electronic part under a counterfeit avoidance clause	LOT	Module 5
Raw material callout requiring traceability	LOT (short text)	Module 2
Item requiring special handling	SH, LOT	Module 5
Explosive material	SH, LOT	Module 5
Radioactive material	SH, LOT	Module 5
DoD serially managed item	IUID, SER	Module 6
Parent item of an embedded item carrying a UII	IUID (parent item text), SER (long text)	Module 6
Warranted serialized item	IUID, SER	Module 6
Special tooling or special test equipment for an MDAP	IUID, SER	Module 6
High risk item identified by the requiring activity	IUID, SER	Module 6

Item or condition	Codes to apply	Taught in
Physical parent of a SER or LOT item, below the end item	SER, TR	Module 2
Installation drawing between a traceable item and the end item	TR (installation text)	Module 2
`End item`	TR (terminator text). Add IUID when the contract requires it: the item text, or the parent item text if the contract names the end item as a parent item. Never SER.	Modules 2 and 6

The matrix gives one row at a time. Real items often hit several rows. Use the method:

1. **List every source that applies**
   Regulation, contract clause, designation, physical condition. Do not stop at the first.

2. **Take the union of the codes**
   Each code once. One LOT covers every source that needs lot records.

3. **Choose the note texts**
   Long or short for SER and LOT. Cycle/usage or age and shelf-life for LL. State the SH control.

4. **Add TR if the item has traceable children**
   Then cascade: every physical parent below the end item takes SER and TR.

Find every source. Apply every code. Then cascade.

CHECK YOURSELF · M8-01

A turbine disk is cycle-limited, is a Critical Safety Item, and is bought under a contract with DFARS 252.211-7003. Its parts list has no children. Which codes does it take?

- [ ] **A** CSI, LL, SER, LOT, IUID, and TR
- [ ] **B** CSI, LL, SER, LOT, and IUID
- [ ] **C** CSI, SER, LOT, and IUID
- [ ] **D** LL and SER

M8-02 Common misreads

Twelve rules trip people up. Each misread below produces a wrong code. The course taught every correct reading. This page gathers them in one place.

The misread	The correct reading	Taught in
A calendar limit makes the item a lot/batch item.	A calendar-limited ALI on a discrete part is still serialized. Calendar time changes the LL note text only.	Module 4
An inspection interval adds LL.	Only a mandatory replacement limit adds LL.	Module 4
A part too small to mark can be lot recorded instead.	The required information moves to the part or its container. The part keeps its unique serial number.	Module 4
The Limit of Validity is a replacement time.	The LOV bounds the engineering data behind the airplane's structural maintenance program. It is not a part replacement time.	Module 4
Specialized metallic fasteners are life-limited.	They have no shelf life. They take LOT under process and batch conditions.	Module 5
Counterfeit avoidance requires serialization.	It requires lot/batch records with a source chain.	Module 5
Beryllium is explosive material.	Beryllium is a toxic hazardous material. It takes SH and LOT under the hazardous material factor.	Module 5
IUID cascades up the tree, like SER.	IUID goes only on items that hold an IUID requirement and on the parent items the contract names.	Module 6
Lot/batch recording can stand in for a UII.	It cannot. Every UII carries a serial number.	Module 6
The end item takes SER.	The contract assigns the end item's identity. It takes TR with the terminator text, and IUID when the contract requires it.	Modules 2 and 6
A Safety-Critical Item is a Critical Safety Item.	The two definitions are independent. A Safety-Critical Item drives no code on its own.	Module 7
A Fracture-Critical Traceable Part is a kind of Fracture-Critical Part.	The two definitions exclude each other. Only the traceable part takes SER on that basis.	Module 7

CHECK YOURSELF · M8-02

Which statements are correct?

Tick all that apply.

- [ ] **A** The end item takes SER when the contract requires IUID.
- [ ] **B** IUID goes only where an IUID requirement or the contract puts it.
- [ ] **C** A calendar-limited ALI on a discrete part takes SER.
- [ ] **D** An inspection interval in the ALS adds LL.
- [ ] **E** A Safety-Critical Item takes no code on that basis alone.
- [ ] **F** Counterfeit avoidance on an electronic part requires a serial number.

M8-03 What reconciliation compares

Take the brake hose again, with the gap fixed. The released parts list says LL and LOT. The build record shows lot 4471. It shows that lot 4471 went into brake line assembly serial BL-0213. That assembly carries a serial number. The chain of records continues.

Reconciliation is that comparison, made for every line. Configuration Management owns it.

EIA-649-1A DEFINITIONS

As-designed configuration. "The configuration of an item as documented by the design activity."

As-built configuration. "The configuration of an item as actually produced. The as-built configuration consists of the as-designed configuration at the time of production as modified with approved engineering changes and variances. In addition, if the as-designed configuration consists of design alternatives, the as-built configuration reflects the actual produced item."

Product structure. "Product structuring is the hierarchical decomposition of a product. Product structures show component relationships, enable visualization of a product in relation to its higher, and lower level components, from any level. When effectively done, product structuring facilitates the 'where used' requirement needed for both traceability and impact analysis."

Configuration status accounting (CSA). "The CM function that formalizes the recording and reporting of the established product configuration information (including historical information), the status of proposed changes, and the implementation of approved changes and changes occurring to product units due to operation and maintenance. CSA implementation includes assurances that the information is current, accurate, and retrievable."

Put the terms together. The part note codes belong to the as-designed configuration. The build records belong to the as-built configuration. Reconciliation compares the two. TR creates the parent-child links in the product structure. Configuration status accounting records the result.

EIA-649-1A asks for the same chain of records. Its status accounting requirements direct the contractor, which the standard calls the Supplier, to do this:

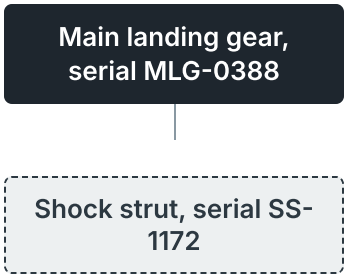
> **EIA-649-1A**
>
> “Record and report all product data inherited from Sub-tier suppliers, to include vendor part numbers and serial numbers, and provide clear traceability from these components to their respective next higher assemblies.”

TR is the code that produces that traceability.

CHECK YOURSELF · M8-03

Build the as-built product structure for one main landing gear. Place each block under its physical parent. Put anything that is not part of the product in the bin.

Write the block each one goes under. Write **BIN** for a block that is not part of this product.



Block	Goes under
Side brace, serial SB-0941	
Brake line fittings, lot F-2210	
Brake hose, lot 4471	
Brake line assembly, serial BL-0213	
Hose material specification	
Side brace lug, forged, lot H-8812	
Shock strut seal set, lot 5530	
Brake line installation work instruction	

M8-04 Two kinds of gap

Reconciliation runs six checks. Each check asks one or two questions.

Check	Question	Typical evidence
`Identity`	Does every SER item have a serial number recorded for each unit? Does every LOT item have a lot/batch number recorded?	Build records, receiving records, supplier certifications
`Link`	Does every recorded item name its next-higher physical parent? Does that parent have a serial number?	Installation records, as-built product structure
`Chain`	Do the links reach the end item without a break? Does the chain of records pass through installation drawings without a serial number?	As-built product structure up to the tail number
`IUID`	Is each UII marked and verified? Is it reported? Is each embedded item reported with the UII of its parent item?	Mark verification records, WAWF and IUID Registry submissions
`Condition`	Does each LL item have the data to start its life status? Is each SH condition recorded with its lot?	Life status records, cure dates, handling records
`Orphan`	Is there a record for every code? Is there a code for every record?	A comparison of the parts list with the records

An orphan in either direction is a finding. A code with no record means the build missed something. A record with no code may mean the design missed something.

Every finding is one of two kinds of gap.

Gap	What it means	Correction	Owner
Design gap	The parts list is missing a code, or carries the wrong code or note text.	Revise the parts list through the change process. Then decide what to do about units already built.	Design Engineering, with Configuration Management
Record gap	The code is right. The build environment did not capture the record.	Capture the record. Correct the process that missed it.	Manufacturing and Quality, with Configuration Management

Do not close a design gap by correcting records alone. The next unit will have the same gap. Do not close a record gap by changing the design. The code was right.

CHECK YOURSELF · M8-04

The parts list shows LOT on a hose. The hose's physical parent carries no code. The build records show the hose lot and no parent serial number. What kind of gap is it, and who corrects it?

- [ ] **A** A record gap. Manufacturing captures the parent serial number by hand.
- [ ] **B** No gap. The record shows the hose lot number.
- [ ] **C** A design gap. Design Engineering adds SER and TR to the parent through change.
- [ ] **D** A record gap. Quality corrects the receiving process.

M8-05 When to reconcile, and the release check

Reconcile at four points.

- **At design release.** Check the parts list against the codes before any record depends on it.
- **During build.** Compare records with codes as the build environment records each unit.
- **Before delivery.** Complete the as-built record up to the end item.
- **After a change** that adds a source, removes a source, or moves an item to a new parent.

The release check

Design release is the cheapest point to catch a gap. Run one check on every parts list before release.

1. **Start at each SER or LOT line**
   Find every line that carries SER or LOT.

2. **Go up one level**
   The physical parent must carry SER and TR.

3. **Keep going up**
   Every physical parent below the end item carries SER and TR. Each installation drawing carries the installation TR text.

4. **Stop at the end item**
   It carries the terminator TR text. It never carries SER.

When an item changes

A modified part may get a new part number. It keeps its serial number. EIA-649-1A states the principle.

EIA-649-1A

Principle CI-6. “When a product is modified, it retains its original product unit identifier, even though its part identifying number is altered to reflect a new configuration.”

Two limits

- **Coverage.** Reconciliation checks codes against records. It cannot find a source that nobody identified.
- **Supplier data.** Reconciliation depends on supplier lot, heat, and serial data. Supply Chain must flow the requirement down. The data must arrive with the part.

CHECK YOURSELF · M8-05

Reconciliation finds a record for every code on a parts list. Can a requirement still be missing?

- [ ] **A** Only if the supplier sent no data.
- [ ] **B** Yes. Reconciliation cannot find a source that nobody identified.
- [ ] **C** No. A full match proves the design is complete.
- [ ] **D** Only on items that carry no SER.

M8-DR Parts-list drill

Read each line. Tick every part note code it takes. Then check it.

STANDING ASSUMPTIONS FOR EVERY LINE

Unless the line says otherwise, the item has no children that carry SER or LOT, it is not the end item, and the contract carries no IUID clause.

A line is right only when the set is exact. A missing code is a defect. An extra code is also a defect. Tick **No code** when nothing applies.

LINE M8-L1 · FIND 3 · PART NO. EX-CMP-DSK-03

DISK, COMPRESSOR, STAGE 3

- Life-limited to 20,000 engine cycles.
- Designated a Critical Safety Item by the design control activity.
- The contract includes DFARS 252.211-7003.

☐ SER ☐ LOT ☐ TR ☐ IUID ☐ LL ☐ SH ☐ CSI ☐ FC ☐ No code

LINE M8-L2 · FIND 5 · PART NO. EX-FTG-PRM-27

FITTING, ATTACH, PREMIUM SELECTED

- The program tests a specimen of each fitting before use to support higher design values.
- Discrete part. ALS entry: replace at 40,000 flight cycles.

☐ SER ☐ LOT ☐ TR ☐ IUID ☐ LL ☐ SH ☐ CSI ☐ FC ☐ No code

LINE M8-L3 · FIND 8 · PART NO. EX-CCA-CTL-61

CIRCUIT CARD ASSY, CONTROL

- Electrostatic discharge sensitive.
- Its children include capacitors that carry LOT.

☐ SER ☐ LOT ☐ TR ☐ IUID ☐ LL ☐ SH ☐ CSI ☐ FC ☐ No code

LINE M8-L4 · FIND 2 · PART NO. EX-LUG-SLP-44

LUG, WING ATTACH, SINGLE LOAD PATH

- A safety-of-flight structural component.
- Single load path.
- Discrete part. ALS entry: inspect every 8,000 flight cycles. No replacement time.

☐ SER ☐ LOT ☐ TR ☐ IUID ☐ LL ☐ SH ☐ CSI ☐ FC ☐ No code

LINE M8-L5 · FIND 10 · PART NO. EX-VLV-SCI-90

VALVE, HYDRAULIC SHUTOFF, SAFETY-CRITICAL

- Identified as a Safety-Critical Item by hazard analysis.
- Its seals carry LOT.
- No other source applies.

☐ SER ☐ LOT ☐ TR ☐ IUID ☐ LL ☐ SH ☐ CSI ☐ FC ☐ No code

M8-EX Exercises

Write your reasoning, not just your conclusion. The reasoning is what is being assessed, and several of these have more than one defensible answer. Where a question turns on a fact you have not been given, **say so, and say what you would ask for and whom you would ask.** Recognizing the missing fact is part of the exercise.

Exercise 1: Code a full parts list

You own the parts list for a main landing gear side brace assembly on a military transport whose contract includes DFARS 252.211-7003. The list has seven items. The side brace is a safe-life PSE with an ALS replacement time of 60,000 landings. The design control activity has designated it a Critical Safety Item. A forged lug detail sits inside the brace. Its raw material heat and lot must be traceable. Bushings are bought under a premium selection, with a coupon tested from each. O-rings carry a shelf-life limit. Specialized fasteners require heat and lot traceability. Synthetic grease carries a shelf-life limit on the lot. A data plate has no requirement. The assembly is installed on the airplane through a landing gear installation drawing.

a. Code every item on the list. For each code, name the source.

b. The side brace hits several matrix rows. Show the union and explain why each code appears once.

c. Which items take LOT? For every one of them except the grease, say whether it takes the long or the short LOT note text, and why.

d. Code the side brace assembly, the installation drawing, and the airplane.

e. Configuration Management finds that the build records show a lot number for the O-rings but no link to the assembly serial number. What is the gap, and what does it cost if the O-ring lot is later found suspect?

Exercise 2: Reconciling an actuator

Configuration Management reconciles pitch trim actuator assembly serial PTA-0057 before delivery. The released parts list shows six lines. The actuator assembly carries SER and TR. A ball screw carries LL and SER. A gearbox subassembly carries no code. A bearing set inside the gearbox carries LOT. A seal kit carries LL and LOT. A motor carries no code. The build record extract shows four entries. Ball screw serial BS-1120, installed in PTA-0057. Bearing set lot 7730, with no parent serial number. Seal kit lot S-2291, received, with no parent serial number. Motor serial M-4410, installed in PTA-0057.

a. Which finding is a design gap? Name the line, the codes it is missing, and who corrects it.

b. The bearing set lot and the seal kit lot both show no parent serial number. Explain why one is a design gap and the other is a record gap.

c. The motor's serial number appears in the build record, but the motor line carries no code. What kind of finding is it, and what do you do with it?

d. Which configuration does the parts list represent, and which does the build record represent? For each of the two gaps in parts a and b, which one do you correct?

e. Units PTA-0050 through PTA-0056 came from the same parts list. What does the design gap mean for them, and when should it have been caught?

Glossary

Airworthiness Limitation Item

ALI

An item holding a mandatory replacement time, structural inspection interval, or related procedure approved as part of the type design and published in the Airworthiness Limitations Section (ALS) of the Instructions for Continued Airworthiness (ICA).

Airworthiness Limitations Section

ALS

The FAA-approved section of the Instructions for Continued Airworthiness. It holds mandatory replacement times, structural inspection intervals, and related procedures.

As-built configuration

“The configuration of an item as actually produced. The as-built configuration consists of the as-designed configuration at the time of production as modified with approved engineering changes and variances. In addition, if the as-designed configuration consists of design alternatives, the as-built configuration reflects the actual produced item.” (EIA-649-1A)

As-designed configuration

“The configuration of an item as documented by the design activity.” (EIA-649-1A)

Build environment

The manufacturing systems and people that build the product and capture its records. Manufacturing and Production owns it.

Cascade

The rule that maintains serialization and traceability records from every traceable item up the product tree to the end item. Every physical parent below the end item takes SER and TR.

Casting factor

An extra factor of safety applied to the design loads of a casting, because a casting’s strength varies more than wrought material does.

Condition code

LL, SH, CSI, or FC. A code that identifies a condition of the item. It creates no record on its own.

Configuration status accounting

CSA

“The CM function that formalizes the recording and reporting of the established product configuration information (including historical information), the status of proposed changes, and the implementation of approved changes and changes occurring to product units due to operation and maintenance. CSA implementation includes assurances that the information is current, accurate, and retrievable.” (EIA-649-1A)

Critical Safety Item

CSI

DFARS 209.270-2 defines an aviation critical safety item as a part, an assembly, installation equipment, launch equipment, recovery equipment, or support equipment for an aircraft or aviation weapon system if the part, assembly, or equipment contains a characteristic any failure, malfunction, or absence of which could cause any of the following: “(1) A catastrophic or critical failure resulting in the loss of or serious damage to the aircraft or weapon system; (2) An unacceptable risk of personal injury or loss of life; or (3) An uncommanded engine shutdown that jeopardizes safety.”

Damage tolerance

damage-tolerant

A design approach in which structure carries load with a crack present until an inspection finds the crack.

Design control activity

For an aviation critical safety item, “the systems command of a military department that is specifically responsible for ensuring the airworthiness of an aviation system or equipment, in which an aviation critical safety item is to be used.” (DFARS 252.209-7010)

Design gap

A reconciliation finding in which the parts list is missing a code or carries the wrong code or note text. Design Engineering corrects it through change.

DoD serially managed item

An item the primary inventory control activity identifies as serially managed under DoDI 8320.04. Critical Safety Items are among them.

Effectivity

“A designation, defining the product range e.g., serial numbers, block numbers, batch numbers, lot numbers, model, dates or event, at which a specific product configuration applies, a change is to be or has been affected, or to which a variance applies.” (EIA-649-1A, adopted from EIA-649)

Electronic part

“An integrated circuit, a discrete electronic component (including, but not limited to, a transistor, capacitor, resistor, or diode), or a circuit assembly.” (DFARS 252.246-7007)

Embedded item

An item inside a parent item that carries its own UII. The contractor reports it with the UII of its parent item.

End item

The top-level asset or system under contract, typically the complete aircraft by tail number or the missile system. It is never an LRU.

Enterprise identifier

“A string of characters that uniquely identifies the enterprise. This identifier is assigned by an issuing organization within a scheme and is unique to that scheme. Example: CAGE CODE, Data Universal Numbering System (DUNS) Number. See also Design Activity Identification. Note: For DoD and NATO, CAGE, and NCAGE, codes are mandatory for use.” (EIA-649-1A)

Find number

The number that ties a parts list line to the same item on the drawing picture.

Fracture-Critical Part

A safety-of-flight structural component that is not single load path and is not sized by durability or damage tolerance requirements, but that requires special emphasis due to the criticality. (MIL-STD-1530D with Change 1)

Fracture-Critical Traceable Part

A safety-of-flight structural component that is either single load path or sized by durability or damage tolerance requirements. (MIL-STD-1530D with Change 1)

Group identifier

“Uniquely identifies a group of like units of the same product manufactured under uniform conditions. Examples: lot number/batch number.” (EIA-649-1A)

Heat of material

One batch of metal melted and poured at one time. All metal from one heat shares one chemistry.

Identity code

SER, LOT, or IUID. A code that gives an item a recorded identity.

Installation drawing

A drawing that depicts installation instructions. It is not consumed into the product as a physical part, so it takes TR and no SER.

Instructions for Continued Airworthiness

`ICA`

The maintenance instructions a design approval holder must prepare under 14 CFR 25.1529. The Airworthiness Limitations Section is part of them.

Life status

“The accumulated cycles, hours, or any other mandatory replacement limit of a life-limited part.” (14 CFR 43.10)

Life-limited part

`life-limited item`

“Any part for which a mandatory replacement limit is specified in the type design, the Instructions for Continued Airworthiness, or the maintenance manual.” (14 CFR 43.10)

Limit of Validity

`LOV`

The limit on the engineering data that supports the airplane’s structural maintenance program. It applies to the airplane. It is not a part replacement time.

Line Replaceable Unit

`LRU`

A unit removed and replaced as a whole on the flight line. It is never the end item.

Link code

TR. The code that links recorded identities into parent-child records.

Lot/batch number

The group identifier for items made from one material lot, process run, or test batch.

Note text

The fixed words that go with a part note code on a parts list. Copy them exactly. Fill only the bracketed fields.

Parent item

“The item assembly, intermediate component, or subassembly that has an embedded item with a unique item identifier or DoD recognized unique identification equivalent.” (DFARS 252.211-7003) Do not confuse it with a physical parent.

Part note code

A short code on a parts list line, such as SER or LOT. It tells the build environment what record to create, or it identifies a condition of the item.

Parts list

The list of items that make up an assembly, with part numbers, quantities, and notes. Part note codes sit on its lines.

Physical parent

The next-higher physical item that holds a traceable item. Installation drawings are not physical parents. Do not confuse it with a parent item, which is an IUID term.

Picture sheet

The drawing sheet that pictures the item. The explosive material note, or a reference to it, goes there.

Premium selection

`premium selected material`

Greater material design values may be used under 14 CFR 25.613(e) “if a ‘premium selection’ of the material is made in which a specimen of each individual item is tested before use to determine that the actual strength properties of that particular item will equal or exceed those used in design.”

Principal Structural Element

`PSE`

A structural element that carries a significant share of flight, ground, or pressurization loads. Its failure could cost the airplane its structural integrity.

Product structure

`product tree`

“Product structuring is the hierarchical decomposition of a product. Product structures show component relationships, enable visualization of a product in relation to its higher, and lower level components, from any level. When effectively done, product structuring facilitates the ‘where used’ requirement needed for both traceability and impact analysis.” (EIA-649-1A) This course calls it the product tree.

Prolongation

Extra material cast or forged onto a part beyond its finished shape. It is cut off and tested as a sample of that part.

Reconciliation

The comparison of the part note codes on the released design with the traceability and product-unit records captured in the build environment. Configuration Management owns it.

Record gap

A reconciliation finding in which the code is right but the build environment did not capture the record. Manufacturing and Quality correct it.

Requirement source

The regulation, contract clause, standard, or engineering basis that creates a traceability requirement for an item.

Safe life

`safe-life`

A design approach for structure that cannot meet the damage-tolerance rules. The applicant shows the structure will not crack during its service life, then retires it before that life ends.

Safe-Life Principal Structural Element

`safe-life PSE`

A PSE for which the applicant establishes that compliance with the damage-tolerance requirements is impractical. Under 14 CFR 25.571(c), it must be shown by analysis supported by test evidence to withstand the repeated loads of variable magnitude expected during its service life without detectable cracks.

Safety-Critical Item

`SCI`

“A hardware or software item that has been determined through analysis to potentially contribute to a hazard with Catastrophic or Critical mishap potential, or that may be implemented to mitigate a hazard with Catastrophic or Critical mishap potential.” (MIL-STD-882E)

Serial number

“A unique number identifying individual units within a series of like items. The serial number does not establish the PN or PIN but tracks the number of items that were produced under the PN or PIN. Serial numbers should be assigned to all items requiring tracking.” (EIA-649-1A)

Single load path

A structural arrangement in which one member carries the load. If that member fails, no other member picks up the load.

Traceable item

An item that carries SER or LOT.

Unique item identifier

`UII`

“A set of data elements marked on items that is globally unique and unambiguous. The term includes a concatenated unique item identifier or a DoD recognized unique identification equivalent.” (DFARS 252.211-7003)

Unit identifier

“A sequentially issued identifier used to designate a specific instance of a product among like products. Examples: Item Unique Identification (IUID), serial number, Vehicle Identification Number (VIN).” (EIA-649-1A)

Wide Area WorkFlow

`WAWF`

The DoD system contractors use to submit receiving reports, including the UIIs of delivered and embedded items.
